



EduConnect: A Web-Based Platform for Academic Collaboration and Resource Sharing

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DECLARATION

This project report is submitted to the Department of Information and Communication Engineering, Noakhali Science and Technology University. I declare that this project report has not been submitted for any kind of degree, diploma or publication.

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ACCEPTANCE

This project report is presented to the Department of Information and Communication Engineering at Noakhali Science and Technology University, Noakhali, as a partial requirement for the completion of the Bachelor of Science (B.Sc.) degree in Information and Communication Engineering.

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I want to thank Allah Ta'ala for giving me the courage and chance to work on my project, which is titled "EduConnect: A Web-Based Platform for Academic Collaboration and Resource Sharing" I would like to express my gratitude to my supervisor, Md. Mahbubul Alam Sir, for all of his creative suggestions, insightful advice, constant inspiration, and steadfast support during this project. Completing my project would have been extremely challenging without his help. I also want to express my gratitude to all of my instructors at the Department of Information and Communication Engineering (ICE) for their wise counsel and help with my courses. Lastly, I express my deepest gratitude to my parents for their love, dreams, and sacrifices throughout my life.

ABSTRACT

A structured academic platform is crucial for minimizing distractions and promoting effective collaboration among students, teachers, and administrators. To address this, EduConnect was developed as a web-based academic environment offering centralized tools for discussion, resource sharing, schedule management, and AI-assisted study support. The platform incorporates features such as class and exam timetable systems, group collaboration spaces, real-time notifications, and user-specific dashboards. Developed using HTML, CSS, JavaScript, PHP, MySQL, and Bootstrap, the system follows an incremental model and includes four key user modules: Admin, Teacher, Student, and Visitor. Research methods involved functional analysis, requirement gathering, and iterative interface design. Academic users were selected as the sample population, and the system's effectiveness was evaluated through testing, feedback, and usability validation. EduConnect ultimately fosters academic engagement, resource accessibility, and collaborative learning by combining educational functionality with an intuitive digital interface.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

A Web-Based Platform for Academic Collaboration and Resource Sharing is a digital solution developed to streamline structured communication and academic engagement among students, teachers, and administrators. It provides a focused and distraction-free environment where users can participate in discussions, share educational materials, and stay updated with academic schedules replacing the fragmented experience of relying on general-purpose social media platforms.

The platform incorporates interactive features such as posts, comments, and reactions to create a familiar and user-friendly experience. However, it intentionally excludes non-academic elements like games, event sharing, or entertainment feeds. This ensures that all interactions within the system remain aligned with academic objectives and contribute to a productive learning ecosystem.

Key functionalities of EduConnect include AI-powered assistance, group study rooms, a smart search system, class and exam schedule displays, and real-time notifications. Built with technologies like HTML, CSS, PHP, and MySQL, the system is responsive and accessible across devices. By combining user-centric design with essential academic tools, EduConnect promotes efficiency, collaboration, and digital advancement within the educational environment.

1.2 Background Study

The concept for EduConnect originated from the growing reliance of students on unstructured social media channels like Facebook for academic discussions. While these platforms offer convenience, they often introduce distractions, reduce focus, and lack the organizational tools needed for effective academic engagement. Existing methods—such as sharing notes through group chats or relying on inconsistent class announcements—were found to be disorganized and inefficient. EduConnect addresses these challenges by offering a distraction-free, feature-rich digital solution focused solely on learning. Key elements like collaborative group spaces, AI tutors, academic

wikis, and two-way notice boards are designed to replicate beneficial communication structures while minimizing off-topic disruptions. The goal is to replace fragmented communication practices with a unified, student-oriented system that enhances productivity and academic interaction.

1.3 Problem Statement

In today's academic environment, students and teachers frequently rely on general-purpose social media platforms—particularly Facebook—for the exchange of academic information. While these platforms enable communication, they also introduce distractions and lack features tailored to educational needs. The absence of a dedicated system for academic collaboration results in inefficiencies such as wasted time, unproductive engagement, and poor resource organization.

The key challenges faced within current systems include:

- a) Distractions caused by non-academic content on social platforms.
- b) Disorganized sharing of notes, resources, and study materials.
- c) Inconsistent delivery of class schedules and exam notification
- d) Fragmented and ineffective communication among students and faculty
- e) Limited tools that meet the structured needs of academic activities
- f) Heavy dependence on platforms not designed for instructional use.

These issues highlight the urgent need for a focused, distraction-free web platform that supports structured communication, timely scheduling, academic content management, and productive collaboration. To address these challenges, this project proposes EduConnect—a web-based academic platform specifically designed to enhance educational interaction, improve information flow, and support students and educators through integrated academic modules and smart collaboration tools. The system emphasizes user-specific access, ensuring that each stakeholder—whether student, teacher, or administrator—interacts with features tailored to their role. By fostering a centralized environment, EduConnect aims to bridge gaps in digital education and promote more effective academic engagement.

1.4 Objectives

The primary goal of this web-based platform is to create a focused and distraction-free academic environment that promotes student engagement, streamlines communication, and offers a suite of tools for effective learning. EduConnect serves as a centralized system for sharing academic content, managing class and exam schedules, receiving updates, and accessing collaborative resources—all tailored to meet the needs of modern learners.

The following are the objectives of the system:

- i. To create a distraction-free platform for academic collaboration.
- ii. To enable sharing of posts, comments, and academic discussions.
- iii. To replace social media with an education-focused system.
- iv. To provide class and exam schedule updates in real time.
- v. To support group study with video rooms and screen sharing.
- vi. To offer AI-based study recommendations and smart search functionality.
- vii. To maintain organized access to class notes and learning materials.
- viii. To allow users to manage personal profiles, schedules, and settings.
- ix. To send instant notifications and alerts for timely academic updates.

EduConnect is a web-based academic platform designed to support students, teachers, and administrators in sharing resources, managing schedules, and collaborating in a focused, distraction-free environment. It includes features like academic posts, group study tools, AI assistance, and real-time notifications to enhance learning and streamline communication.

1.5 Motivation of the work

The motivation for developing EduConnect arises from the need to reduce distractions and time wastage caused by using general social media platforms like Facebook for academic purposes. While such platforms support communication, they often divert students from their studies due to non-academic content. EduConnect is designed to create a focused, education-oriented environment that enhances learning

through features like AI-powered study assistance, group discussions, exam alerts, class schedules, and centralized sharing of study materials. Tools such as group video study rooms and personalized dashboards further support organization and collaboration. The platform aims to replace inefficient practices with a system that encourages productivity, academic engagement, and structured knowledge sharing. Moreover, many students face difficulties accessing class notes, schedules, and updates in an organized way. EduConnect solves this by offering a centralized platform, reducing reliance on informal channels and ensuring reliable, secure academic communication.

1.6 Scope of the Study

This project aims to develop a web-based platform that enhances collaborative learning among students while providing a distraction-free environment for academic resource sharing. The system is designed to support core academic functionalities, such as real-time communication, intelligent study tools, and organized information access. Key features include group video study rooms, AI-powered tutoring, exam and schedule notifications, and a structured note-sharing system.

- i. Develop a web platform for academic discussions and collaboration.
- ii. Enable group video study rooms with whiteboard and screen sharing.
- iii. Integrate AI for study suggestions, search, and exam support.
- iv. Provide notifications for exams, class schedules, and semester updates.
- v. Allow users to share and access class notes by subject and term.
- vi. Support profile, password, and calendar management.
- vii. Reduce academic dependence on social media through a focused system.

This project focuses on building a web-based platform to enhance academic collaboration among students in a distraction-free environment. It includes features such as group video study rooms, AI-based learning support, real-time alerts, and organized note sharing. The system allows users to manage their profiles and schedules while promoting effective academic communication without relying on social media. Developed with HTML, CSS, and PHP, the platform is accessible, user-friendly, and scalable for future academic use.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

In order to understand the information and concepts pertaining to web-based academic collaboration platforms it is important to read published work done by experts in the field. The purposes of this review are to review, examine, and gain knowledge regarding the design and operations of online systems that support educational communication and exchange of resources. This paper aims at discussing the principles of the web-based academic platform development and presents the main arguments of the scholars and practitioners of related disciplines. It relies with systematic literature review on similar known systems such as educational networking platforms and resource sharing tools to create ground for the proposed project.

2.2 Related Works

Several researchers have examined the impact of digital and social platforms on academic performance. The following literature highlights the need for structured academic environments that minimize distractions and enhance learning through focused, collaborative tools.

Academic platforms designed for focused learning, as discussed by Rosen et al. [1] in 2013 help reduce distractions and enhance students' academic performance by providing an environment tailored to effective learning.

According to Manca and Ranieri [2] in 2016, the features of Computer-Mediated Communication (CMC) should be integrated with academic resources in educational platforms. Their findings emphasized that well-designed systems effectively capture students' attention while minimizing distractions.

Greenhow et al. [3] in 2009 found that although students used social media for collaborative learning, they were often distracted. The study recommended the development of dedicated academic platforms to mitigate these distractions.

According to Selwyn [4] who explored the role of Social Networking Sites (SNS) in education, such platforms tend to divert learners from their academic objectives. He suggested that purpose-built educational platforms can better support collaborative learning by reducing complexity and unnecessary features in 2009.

Junco et al. [5] in 2011 revealed that frequent use of Facebook among university students leads to increased distractions and reduced academic performance. This underscores the importance of structured academic platforms that encourage productive collaboration.

Surowiecki [6] in 2005 take a look at collective intelligence and stressed that having shared stockpiling enhances collective learning. In enabling students to post information and seek information, platforms increase the scope of knowledge disbursement.

Pea [7] in 1993 reviewed how educational technology reduces cognitive load in the works flow review process managing the works flow tools such as class schedules and examination notification aids performance.

Kirschner and Karpinski [8] Studies show a negative correlation between social network use, especially Facebook, and student grades, highlighting how unstructured academic tools can become distractions in 2010.

Dabbagh and Kitsantas [9] in 2012 explored how Personal Learning Environments (PLEs) supported by social media can enhance self-regulated learning. However, they emphasized that without proper structure and guidance, these tools may overwhelm learners instead of supporting effective education.

Hrastinski and Aghaee [10] in 2012 examined student preferences in online learning environments and found that while learners valued interaction, they preferred platforms specifically designed for academic tasks over general-purpose social media, due to better alignment with their study goals.

Roblyer et al. [11] in 2010 found that while students were open to using social media for learning, faculty were concerned about its lack of structure and distractions. They highlighted the need for dedicated academic platforms aligned with curriculum goals.

Madge et al. [12] in 2009 investigated Facebook use among university students and found that although it supported social bonding, students were reluctant to use it for academic purposes due to its informal and distracting nature.

Veletsianos and Navarrete [13] in 2012 highlighted that while emerging technologies promote open and flexible learning, the absence of academic-oriented structure often leads to distractions and reduced focus, calling for platforms tailored to academic engagement.

Cain and PolICASTRI [14] in 2011 studied pharmacy students' use of Facebook and concluded that the social nature of the platform often conflicted with academic needs. They recommended separating social and academic interactions through dedicated platforms to ensure better learning outcomes.

According to Redecker et al. [15] in 2011, Socialmedia can aid learning and collaboration, but lacks structure. Researchers suggest creating academic platforms with social media features, including organized discussions and resource sharing.

Bosch [16] found that although Facebook facilitated student communication, it lacked academic structure and contributed to distraction. The study advocated for alternative platforms designed specifically to meet educational needs in 2009.

The reviewed studies highlight that while social media platforms support communication, they often distract students and reduce academic performance. Researchers agree that structured, purpose-built academic platforms are more effective for focused learning and collaboration. Therefore, developing a platform like EduConnect aligns with the need for a secure, organized, and distraction-free academic environment.

CHAPTER 3

METHODOLOGY

3.1 Introduction

The software development methodology is a process or collection of processes used in software development. This is the most important part of any project development. In this section, I will discuss the development process briefly of my system.

The project is divided in 4 modules.

- A.** Admin module
- B.** Student module
- C.** Teacher module
- D.** Visitor module

3.1.1 Admin Module

Admin module will have the following elements:

- i. Admin Dashboard
- ii. User Management Panel
- iii. Class Schedule Management
- iv. Exam Schedule Management
- v. Notice Management Panel
- vi. Academic Syllabus Management

The Admin Module provides centralized control for managing the platform. It allows administrators to approve new users, create and update class and exam schedules, publish academic notices, and manage the academic syllabus. This module ensures smooth coordination and efficient administration across all user levels. It also helps maintain data accuracy and platform integrity by allowing real-time updates and monitoring.

3.1.2 Student Module

Student module will have the following elements:

- i. Home Dashboard
- ii. Registration and Login System
- iii. Class Schedule Viewer
- iv. Exam Schedule Viewer
- v. Academic Interaction Hub
- vi. Note Sharing Panel
- vii. Study Materials Explorer
- viii. Smart User Finder
- ix. Group Management System
- x. Private Notes Archive
- xi. Smart Task Tracker
- xii. Notification Panel
- xiii. Unified Profile Hub
- xiv. View User Profile

The Student Module offers access to schedules, study materials, note sharing, group interactions, and task tracking, all through a user-friendly dashboard.

3.1.3 Teacher Module

Teacher module will have the following elements:

- i. Home Dashboard
- ii. Registration and Login System
- iii. Class Schedule Management
- iv. Exam Schedule Management
- v. Resource Sharing Panel
- vi. Newsfeed Panel
- vii. Campus View Panel

The Teacher Module enables teachers to manage class and exam schedules, share resources, and interact with students through a dedicated dashboard.

3.1.4 Visitor Module

Visitor module will have the following elements:

- i. Student Dashboard
- ii. Notice Board Overview
- iii. AI Academic Assistant Module
- iv. Contact & Support Form

The Visitor Module allows users to access notices, syllabus, routines, and results without login. It also includes AI academic assistance and a contact form for support.

3.2 Requirement Specification

Requirement Specification provides brief explanation about the goals that a system, product or project should pursue, taking into account both functional and non-functional requirements.

3.2.1 Functional Requirement

These define the services the system must provide, how it should handle inputs, and how it should behave in specific situations:

- i. **Register / Sign Up**
Students, teachers, and admins can register by providing required details.
- ii. **Authentication**
System validates login credentials before granting access.
- iii. **Login**
Users log in with email and password to access their dashboard.
- iv. **Profile Update**
Users can update personal details from their dashboard.
- v. **Class & Exam Schedule Management**
Admins and teachers can create or edit schedules; students can view them.
- vi. **Post Creation and Interaction**
Users can create posts, react, comment, and share academic content.

- vii. **Note Sharing**
Users can upload and access notes based on course and semester.
- viii. **Study Material Search**
Allows filtering and browsing of academic content.
- ix. **Group Management**
Users can create, join, and manage academic discussion groups.
- x. **Group Forum**
Group members can post and interact in private forums.
- xi. **Private Notes**
Users can upload and manage personal study materials.
- xii. **Task Tracker**
Users can add and track academic or personal tasks.
- xiii. **Notifications**
Real-time alerts for schedules, posts, and updates.
- xiv. **User Finder**
Search for users by name, batch, location, or blood group.
- xv. **AI Assistant**
Users can get academic help through the AI chat interface.
- xvi. **Contact & Feedback**
Visitors and users can contact admin or submit feedback.

3.2.2 Non-Functional Requirements

Non-functional requirements are those that outline the qualities, restrictions, and requirements that a system or product must possess in addition to its primary functionality.

- i. **Usability Requirement**
The system should be easy to use on both desktop and mobile, with a clean and intuitive interface requiring no special training.
- ii. **Availability Requirement**
The platform must be accessible 24/7 throughout the academic year, ensuring no unexpected downtime.

iii. **Efficiency Requirement**

In case of system failure, recovery must be completed within one hour to minimize disruptions.

iv. **Accuracy Requirement**

All academic data, including schedules and notes, must be updated in real time with high accuracy and consistency.

v. **Performance Requirement**

The system should respond to user actions within 2 seconds, with up to 5 seconds allowed for heavier operations.

vi. **Error Handling**

The system should handle both expected and unexpected errors smoothly while protecting data and guiding the user.

vii. **Security Requirement**

Sensitive data and administrative functions should only be accessible to authorized users, with role-based access control in place.

3.3 **Process Model**

A programming process model is a model which is an abstract depiction of a method from a specific viewpoint. There are a variety of broad software process models, including the waterfall model, evolutionary development, formal systems development, and reuse-based development, among others. This research will view the following four models:

- i. Waterfall model
- ii. Iteration model
- iii. Spiral model
- iv. Incremental Model

3.3.1 **Waterfall Model**

The Waterfall model is similar to a step-by-step process for developing computer software. In this model, you must complete one phase before moving on to the next, similar to how water flows down a waterfall. So, it's a loop where you go through each step one after the other until you're finished [17].

3.3.2 Iterative Model

This model is based on a loop in the software development process. It starts with creating a small version of the software that takes into account all of the steps. Then, in each round, more elements, features are added. Each time this loop is completed [18].

3.3.3 Spiral Model

The Spiral Model is a risk-driven software development approach combining prototyping and the Waterfall model. Each loop represents a development phase, with risk evaluation being a key focus throughout [19].

3.3.4 Incremental Process Model

The Incremental Model breaks the project into smaller parts, developed in cycles with planning, building, and testing for each. It delivers usable software early and adds features over time. This model suits the project due to its structured, step-by-step approach [20].

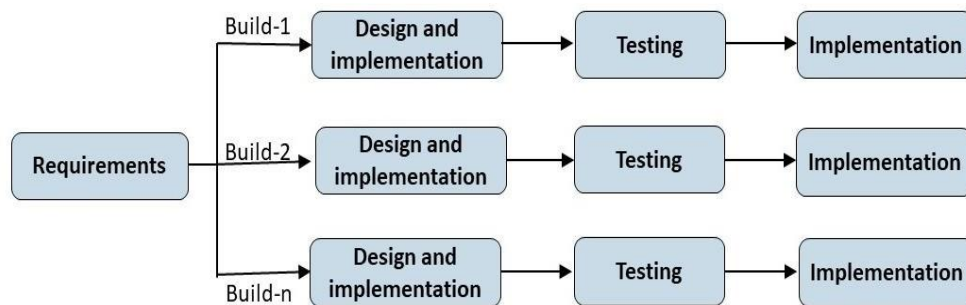


Figure 3. 1: Incremental life cycle model

The Reason Why I used Incremental Process Model:

- a) The full system's requirements are well-defined and comprehended.
- b) Primary requirements are specified, though certain details may evolve over time.
- c) There's a necessity to launch the product in the market timely.
- d) Integration of new technology is underway.

- e) Resources with needed skill set are not available.
- f) There are specific high-risk features and objectives.

3.4 System Design

The system is design depends on the problem analysis and the requirements analysis. Based on this, I will try to design this system in such a way that will give users faster execution of the system.

The system design consists of 2 parts:

- a) Use case diagram design
- b) E-R diagram

Based on these designs, the system is developed using the selected programming languages and tools. This process transforms the design model into a functional web application. In the EduConnect academic platform, the system is divided into four main modules: Admin, Teacher, Student, and Visitor. A centralized user database is used to identify the role of each user and redirect them to their respective module. Each module provides role-specific features such as schedule management, academic collaboration, resource sharing, and profile handling to ensure a smooth and secure user experience.

I will use the following tools to design the Use-case diagram and E-R diagram.

Table 3.1 : Tools used for Use-Case Diagram, E-R diagram Design

Use-Case Diagram Designing Tool	Edrawmax
E-R Diagram Designing Tool	Draw.io

3.4.1 Use-Case Diagram Design

The use case diagram illustrates how different users interact with the system, highlighting the specific features and functionalities each user can access and operate within the platform.

The user interactions and system functionalities are illustrated in the following Use Case Diagram:

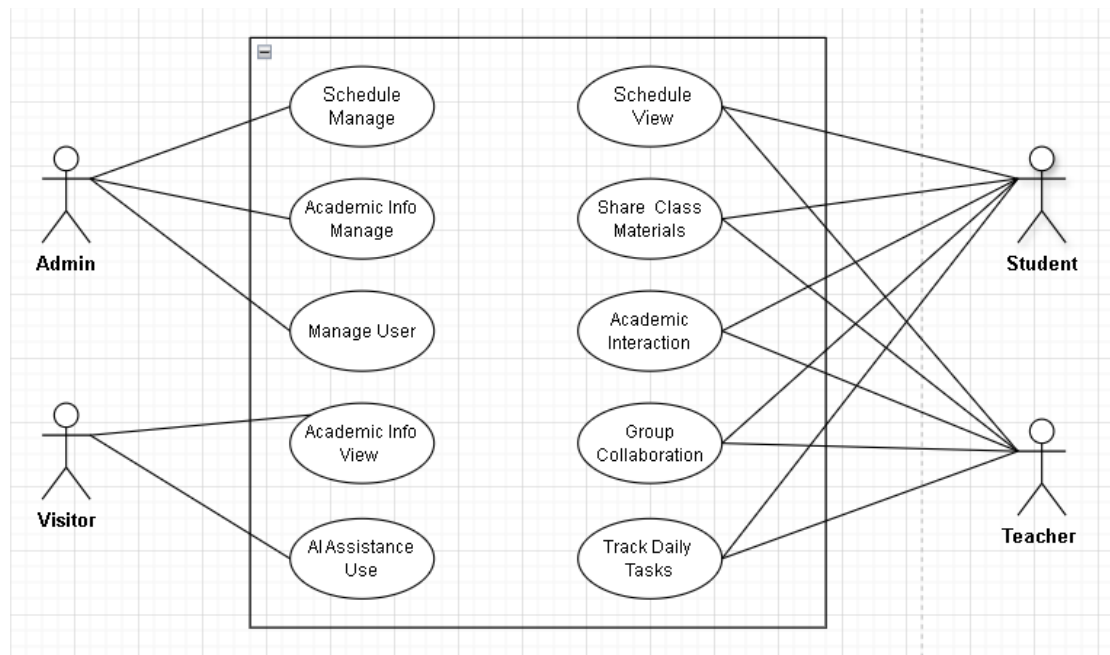


Figure 3. 2: Use case diagram for the system

The use case diagram illustrates how different users interact with the EduConnect system based on their roles. Admins manage users, schedules, and academic content; teachers and students can view schedules, share materials, collaborate in groups, and engage in academic discussions; while visitors have limited access, such as viewing public academic info and using AI tools. Each role is linked to specific functionalities to ensure a focused and organized academic environment. This visual representation helps developers and stakeholders understand system behavior clearly, defines role-based access control, and ensures all core interactions are addressed during the design and development phases.

3.4.2 E-R Diagram Design

The E-R diagram illustrates how entities and their associated attributes are interconnected within the system. It visually represents the structure of the database by showing the relationships between different entities and how they interact with each other.

The entity-relation of the database model is shown by the following E-R diagram:

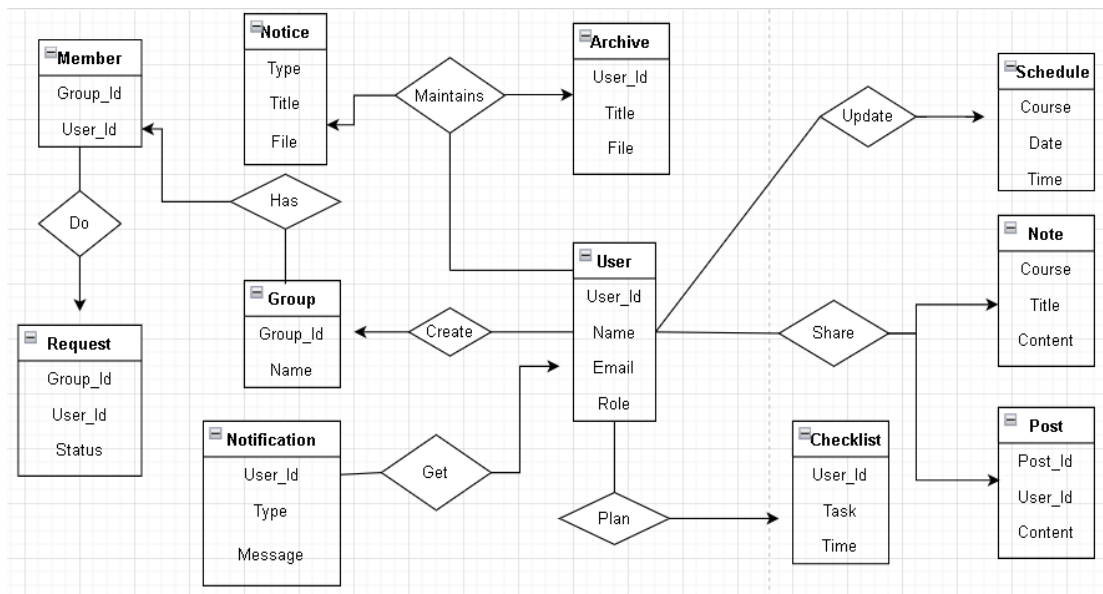


Figure 3. 3: E-R diagram for the system

This ER diagram outlines key entities in an academic platform. Users can create groups, share posts, notes, and archives, and manage schedules and checklists. Group members can request to join, receive notifications, and access notices; all relationships are mapped through user roles and interactions.

3.5 Database Design

The creation of a comprehensive data model for a database is referred to as database design. This logical data model includes all logical and physical design elements, along with the physical storage attributes necessary to develop a design using a data definition language, which can then be utilized to construct a database [21].

For this project, it involved identifying entities like students, teachers, and admins, along with their relationships, to build a structured data model. This model defines how data is stored, accessed, and managed using tables, forms, and queries within the DBMS, ensuring smooth interaction and functionality across all modules of the web application. The database used in my project is mentioned below, which ensures organized and efficient data management throughout the system.

3.5.1 User Registration Table

id	user_name	password	email	reg_no	role	batch_no
1	Ajj	12345	ajjj@gmail.com	2011023	student	15
14	Tarifa	12345	tarifa@gmail.com	2011016	student	16
16	munna	12345	munna@gmail.com	2011070	student	17
19	samapan	12345	samapan@gmail.com	2011024	student	15
20	sajib	12345	sajib@gmail.com	2011031	student	15
23	Naeem	12345	nayemsir@gmail.com		teacher	

Figure 3.4: User Registration Table

3.5.2 Class Schedule Table

date	course_name	start_time	end_time	venue	id	batch
2025-07-13	Digital Logic Design	8:15	9:0	802	55	15
2025-07-13	Digital Logic Design Lab	10:0	11:0	805	56	19
2025-07-13	Optical Communication	15:15	16:0	802	57	15
2025-07-14	Digital Logic Design	14:0	15:0	802	58	18

Figure 3.5: Class Schedule Table

3.5.3 Exam Schedule Table

id	exam_name	date	batch
1	DSA Lab Exam	2025-07-24	16
2	Defense	2025-07-31	15
3	Math CT	2025-07-16	17
4	C++ Exam	2025-07-26	18
5	Multimedia Exam	2025-07-31	16

Figure 3.6: Exam Schedule Table

3.5.4 User Post Table

id	user_id	content	created_at	file_path	privacy
1	7	DBMS Note	2025-06-23 20:10:56	1750692511_project_____22.pdf	15
2	8	Hi everyone	2025-06-23 20:11:44	133333333.jpg	public
4	7	Hi 16	2025-06-23 20:40:22	NULL	16
6	7	This is my resume	2025-06-23 21:14:45	1750691685_Ajij.pdf	private

Figure 3.7: User Post Table

3.5.5 Class Note Table

id	course_name	semester	author	title	type	file	uploaded_date
40	Microwave Engineering	3/1	Mohammad Ajij	Chapter 2	Reff. Book	1751137381_Ajij.docx	2025-06-29 01:03:02
41	Digital Logic Design	2/1	Mohammad Ajij	Lec-2	Other	1751739675_Ajij1.pdf	2025-07-06 00:21:15
46	Optical Communication	4/2	Mahbubul Alam Naeem	Lec-8	Lecture	1752505298_gdgdg.cpp	2025-07-14 21:01:38

Figure 3.8: Class Note Table

3.5.6 Available Groups Table

id	name	cover_image	created_by	created_at
30	ICE 15 Batch	uploads/1751796173_WhatsApp Image 2025-07-06 at 3....	18	2025-07-06 16:02:53
31	For Machine Learner	uploads/1752430924_what-is-machine-learning-1024x5...	18	2025-07-14 00:22:04
32	For Programmer	uploads/1752431010_competitive programming.jpeg	19	2025-07-14 00:23:30

Figure 3.9: Available Groups Table

3.5.7 Group Post Table

id	group_id	user_id	content	created_at	file_path
49	26	7	define stack	2025-06-28 10:40:18	
56	24	7	What is a stack?	2025-07-06 16:50:15	
57	32	19	hey	2025-07-15 11:42:37	1752558157_2.jpg

Figure 3.10: Group Post Table

3.5.8 Group Members Table

id	group_id	user_id	joined_at
99	30	18	2025-07-06 16:02:53
100	30	4	2025-07-06 16:02:53
104	30	11	2025-07-06 16:02:53

Figure 3.11: Group Members Table

3.5.9 Academic Syllabus Table

id	si_no	course_code	1	course_title	credit	course_type	year_term
13	7	ICE 4218		Project and Thesis	4	Compulsory	Year-4 Term-II
17	11	ICE 4209		Geographical Information System	3	Optional	Year-4 Term-II
16	10	ICE 4208		Digital Image Processing Lab	1	Optional	Year-4 Term-II

Figure 3.12: Academic Syllabus Table

3.5.10 Notifications Table

id	user_id	type	reference_id	message	is_read	created_at
1	1	new_note	0	A new note has been shared.	0	2025-06-28 21:56:22
2	2	new_note	0	A new note has been shared.	0	2025-06-28 21:56:22
3	3	new_note	0	A new note has been shared.	0	2025-06-28 21:56:22
4	4	new_note	0	A new note has been shared.	0	2025-06-28 21:56:22

Figure 3.13: Notifications Table

3.5.11 Notices Table

id	type	title	file_url	uploaded_date
1	Re Admission 2023-24	Re Admission		2025-06-19
2	Re Admission 2023-24	Re Admission		2025-06-19
7	Exam form fillup	Exam	uploads/notices/1751790184_Cover_page.pdf	2025-07-06
8	Others	Scholarship	uploads/notices/1751800964_Cover_page.pdf	2025-07-06

Figure 3.14: Notices Table

3.5.12 Personal Note Table

id	title	filename	uploaded_at
11	lab	Multimedia/1750618798_project_____2.pdf	2025-06-23 00:59:58
13	TTheory	DBMS/1750619035_project_____2.pdf	2025-06-23 01:03:55
16	Inheritance Topic	OOP/1751824976_os_ct(3).pdf	2025-07-07 00:02:56

Figure 3.15: Personal Note Table

3.5.13 Personal Task Lists Table

id	task	date	time	completed	user_id
28	Learn DP	2025-07-14	00:01:07	1	18
29	Learn JS	2025-07-14	00:02:08	0	18
30	Practice Graph	2025-07-13	00:02:30	1	18

Figure 3.16: Personal Task Lists Table

3.5.14 Teacher Class Schedule Table

id	teacher_name	course_name	semester	date	start_time	end_time	venue	created_at	teacher_id
6	Sabbir Sir	Digital Logic Design	2/1	2025-07-14	14:00:00	14:00:00	802	2025-07-14 17:33:17	1
7	Naeem Sir	Multimedia Communication	4/2	2025-07-14	07:00:00	10:00:00	802	2025-07-14 21:08:13	23

Figure 3.17: Teacher Class Schedule Table

3.5.15 Teacher Exam Schedule Table

id	exam_name	date	batch	teacher_id
1	DSA Lab Exam	2025-07-24	16	18
2	Defense	2025-07-31	15	18
3	Math CT	2025-07-16	17	23
4	C++ Exam	2025-07-26	18	0

Figure 3.18: Teacher Exam Schedule Table

3.6 System Implementation

This process includes figuring out how the information system should look, making sure it works properly, and checking that it meets quality standards. It covers building the system, making it work, and ensuring it's of good quality.

3.6.1 Software Requirements

The following software will be used to complete the project:

Table 3. 2: Software Requirement of the system

Operating system	Linux, Windows XP or later
Browser	Google Chrome, Firefox
Editor	Notepad++, Visual Studio Code, Sublime Text or any text editor.

3.6.2 Hardware Requirements

The following hardware will be used to complete the project:

Table 3. 3 : Hardware Requirements of the System

Processor	Standard Processor
RAM	2GB RAM or more
Hard Disk	50 GB or more
Keyboard, Monitor, Mouse	Standard

3.7 Technologies and Tools

Basically, LAMP stack will be used for this project. In this client-server architecture, there are two categories of tools for creating this project. To develop this software project, I am going to use the following technologies.

3.7.1 Technologies for Client-Side Scripting

- i. HTML (Hypertext Markup Language)
- ii. CSS (Cascading Style Sheets)
- iii. Bootstrap (CSS Framework)
- iv. JavaScript

3.7.2 Technologies Server-Side Scripting

- i. PHP (Hypertext Preprocessor)

3.7.3 Hosting and version control system

- i. Heroku (for free hosting)
- ii. GIT (for version control system)
- iii. XAMPP (for development and testing PHP applications)

3.7.4 Database Related Tools

- i. Mysql

(i) HTML:

HTML is the core language of the web, structuring and organizing online content. "Hypertext" refers to links connecting web pages across sites. While HTML builds the content, CSS styles it, and JavaScript adds interactivity. HTML enables sharing and linking content across the World Wide Web [22].

(ii) CSS:

CSS (Cascading Style Sheets) is a simple method for improving the visual aspects of web documents, such as fonts, colors, and spacing [23]. This stylesheet language is used to define the looks of HTML or XML documents by specifying how elements should be displayed across different mediums such as screens, print, speech, and other forms of media [24].

(iii) Bootstrap:

Bootstrap is an open-source CSS framework for developing responsive, mobile-first front-end web applications. It provides CSS-based design templates for components such as typography, forms, buttons, navigation, and various interface components, with the capability to include JavaScript [25].

(iv) Java Script:

JavaScript is a programming language that adds interactivity to websites. It makes static pages dynamic, enhancing user experience. Used in browsers, servers, and apps for actions like clicks, animations, and timers. [26].

(v) PHP (Hypertext Preprocessor):

PHP is a server-side scripting language ideal for dynamic, database-driven websites. It works efficiently with MySQL to handle CRUD operations and backend tasks. Supports user authentication, data submission, and complex queries. Connects to MySQL via native extensions like MySQL or PDO [27]

(vi) MySQL:

MySQL is a popular open-source RDBMS developed by Oracle, known for reliability, scalability, and performance. It uses SQL for data access and supports ACID transactions, making it ideal for web applications. Available under GPL and commercial licenses [28].

(vii) Git:

Git is an open-source distributed version control system designed to help developers manage and track changes to their code over time. Unlike older centralized version control systems, Git is distributed, allowing each developer to have a full copy of the code base on their local machine [29].

3.8 Testing

The project is extensively tested across multiple contexts, containing disparate browsers, operating systems, hardware, and data. Making ensuring the product satisfies user expectations and technical requirements is the aim of software testing. The software modules are subjected to testing. Software testing includes both validation and proofing. There are two methods used in software testing [30].

3.9 Conclusion

This chapter presented the step-by-step methodology for developing EduConnect, covering system design, tools, technologies, and testing. Using the Incremental Model, the platform was built module-wise with a focus on functionality, usability, and reliability.

CHAPTER 4

RESULT AND DISCUSSION

After implementing the academic web application titled “Web-Based Platform for Academic Collaboration and Resource Sharing,” the system was structured into several integrated module– Student module, Teacher module, Visitor module and Admin module. The outcomes of this academic web application project are outlined below:

4.1 Student Module

The Student Module provides students with access to key features like the dashboard, class schedules, notes, study materials, and group interactions to manage their academic activities efficiently.

4.1.1 Home Page

The dashboard appears when a user visits the platform's main URL.

4.1.1.1 Top Navigation Bar

Provides access to the following sections:

- i. Home
- ii. Profile
- iii. Group
- iv. Notifications
- v. Dashboard
- vi. Logout

It also includes a user search bar for quickly finding users or posts.

4.1.1.2 Main Dashboard Sections

The Main Dashboard Consists of Four Key Sections for Streamlined Academic Management:

i. Class Schedule Panel

Displays the current day's classes along with time and room details. Users can click the "Update" button to refresh the schedule.

ii. Newsfeed Panel

Allows users to post text or upload files. Users can also interact with posts through like, dislike, and comment options.

iii. Exam Panel

Lists upcoming exams with date and title. Includes an "Update" button for modifications.

iv. Note Sharing Panel

Enables users to browse existing class notes or upload and share their own.

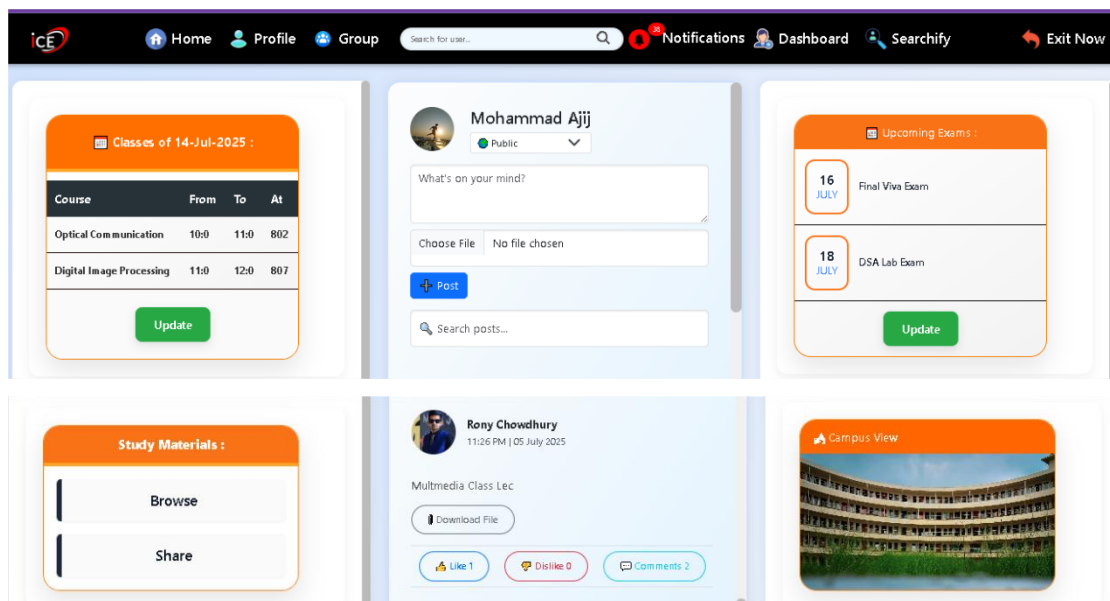
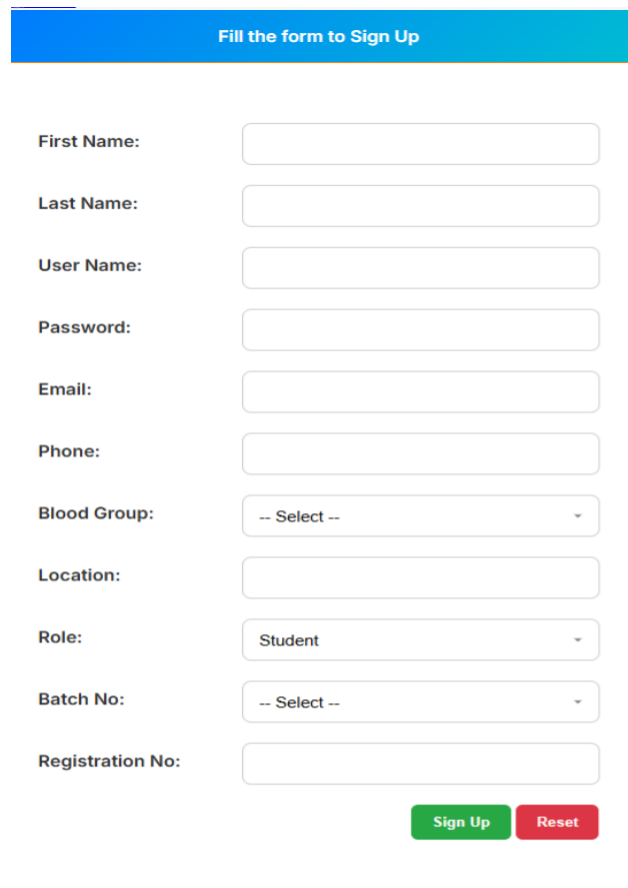


Figure 4.1: Home Page

4.1.2 Registration Page

This registration form allows new users to create an account by submitting personal, academic, and contact details. After signing up, the request goes to the admin for approval. Once approved, users can log in and access all platform features.

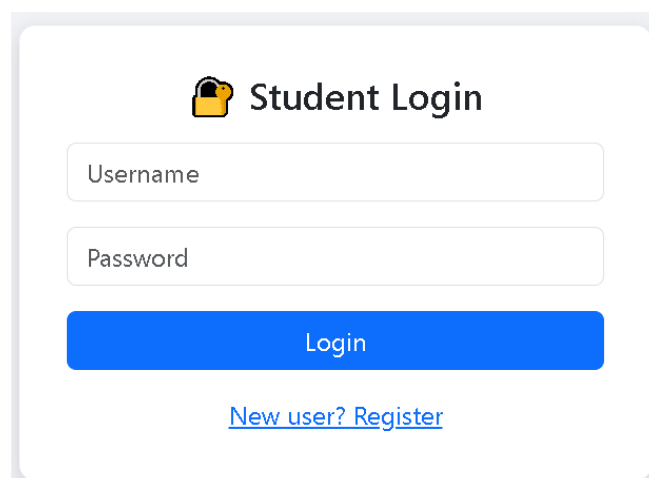


The registration form is titled "Fill the form to Sign Up" in a blue header. It contains the following fields: First Name, Last Name, User Name, Password, Email, Phone, Blood Group (dropdown menu with "-- Select --"), Location, Role (dropdown menu with "Student"), Batch No (dropdown menu with "-- Select --"), and Registration No. At the bottom right, there are two buttons: "Sign Up" (green) and "Reset" (red).

Figure 4.2: Registration Page

4.1.3 Login Page

After registration, user can login their account by inserting valid email address and password. The system also includes options for password recovery and keeps the login process secure to protect user information.



The login page is titled "Student Login" with a lock icon. It features two input fields: "Username" and "Password". Below these fields is a blue "Login" button. At the bottom, there is a link that says "New user? Register".

Figure 4.3: Login Page

4.1.4 Academic Interaction Hub

This module provides a centralized space for users to share academic posts, ask questions, upload files, and engage with others through comments and reactions. It fosters collaborative learning and communication within the platform. Users can easily contribute content, participate in discussions, and access valuable resources shared by peers or teachers, enhancing overall academic interaction.

4.1.4.1 Post Creation Box Page

Users use this section to share thoughts or academic content by typing a message and uploading files. Privacy can be managed by selecting the desired batch or section, allowing posts to be shared only with specific groups.

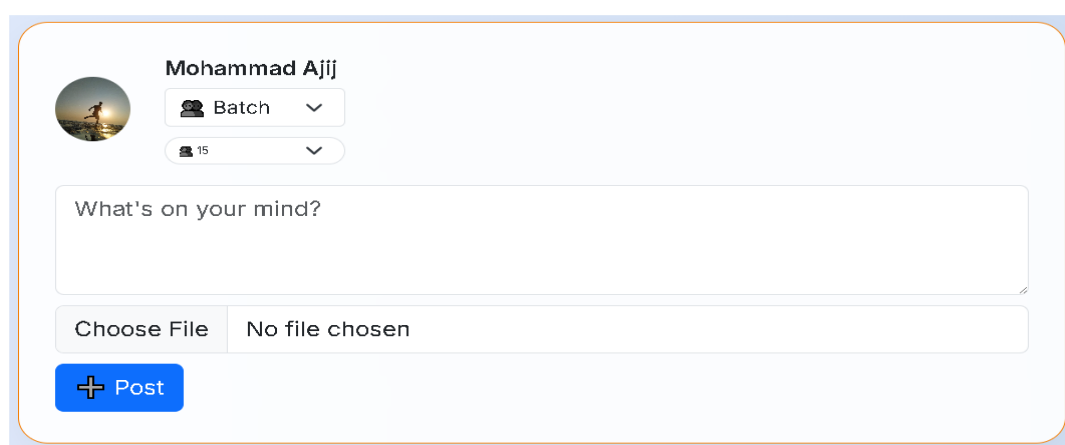
The image shows a user interface for creating a post. At the top left is a circular profile picture of a person. To its right is the name 'Mohammad Ajij'. Below the name are two dropdown menus: the first is labeled 'Batch' with a person icon, and the second is labeled '15' with a document icon. Below these is a large text input area with the placeholder text 'What's on your mind?'. Under the text area is a file upload section with a 'Choose File' button and the text 'No file chosen'. At the bottom left of the form is a blue button with a plus icon and the word 'Post'.

Figure 4.4: Post Creation Box

4.1.4.2 Shared Post with Comment Interaction Panel

This page displays a shared post along with options to like, dislike, and comment, enabling interactive engagement among users. Students can view and write comments, allowing for quick feedback and discussion. The “Download File” button ensures easy access to shared class materials or updates, streamlining resource distribution. These features collectively promote collaborative learning, real-time communication, and a more connected academic environment.

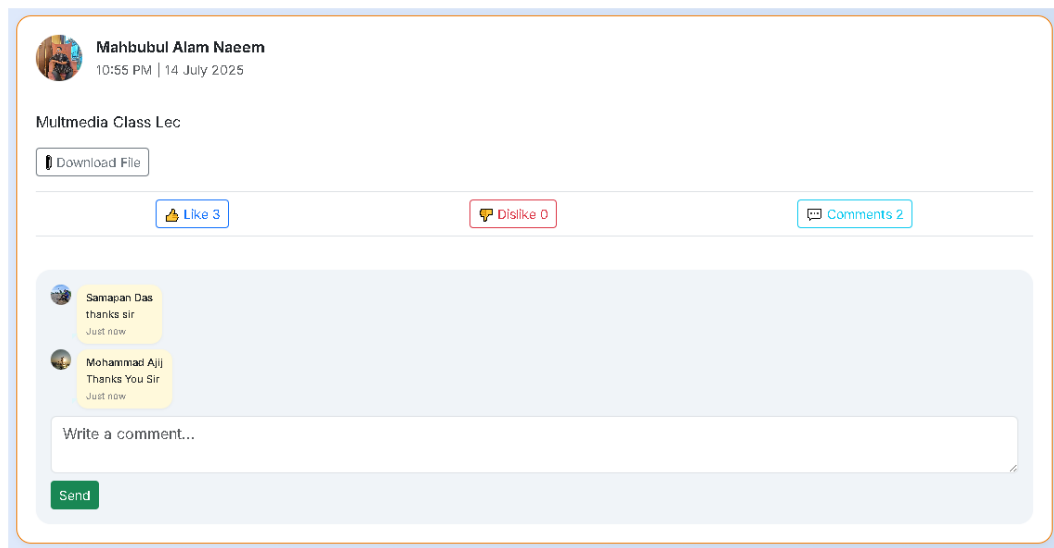


Figure 4.5: Shared Post with Comment Feature

4.1.5 Class Schedule Page

Users can see this type of class schedule to stay informed about their daily academic sessions, including course timing and room allocation. It ensures timely attendance and better class management.

Classes of 06-Jul-2025 :			
Course	From	To	At
Digital Logic Design	8:15	9:0	802
Digital Logic Design Lab	10:0	11:0	805

Update

Figure 4.6: Class Schedule Page

4.1.6 Exam Schedule Page

Users can see this type of exam schedule page to stay updated about their upcoming exams. It helps them manage their preparation and plan accordingly.

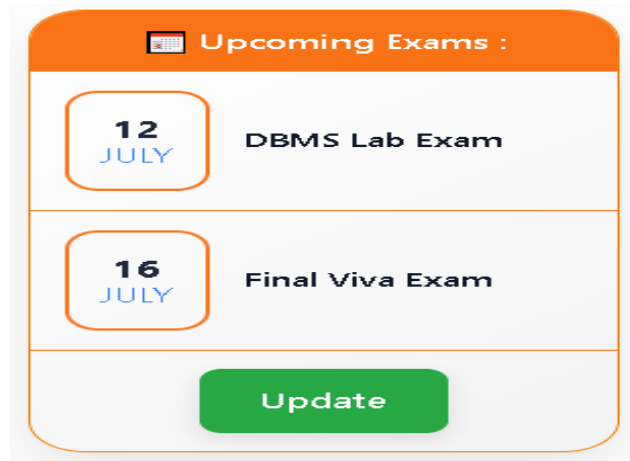


Figure 4.7: Exam Schedule Page

4.1.7 Smart User Finder Module

This module helps users quickly find others by filtering based on name, blood group, location, and batch, making academic connection and collaboration easier. It enhances networking opportunities by providing accurate and fast search results tailored to user needs.

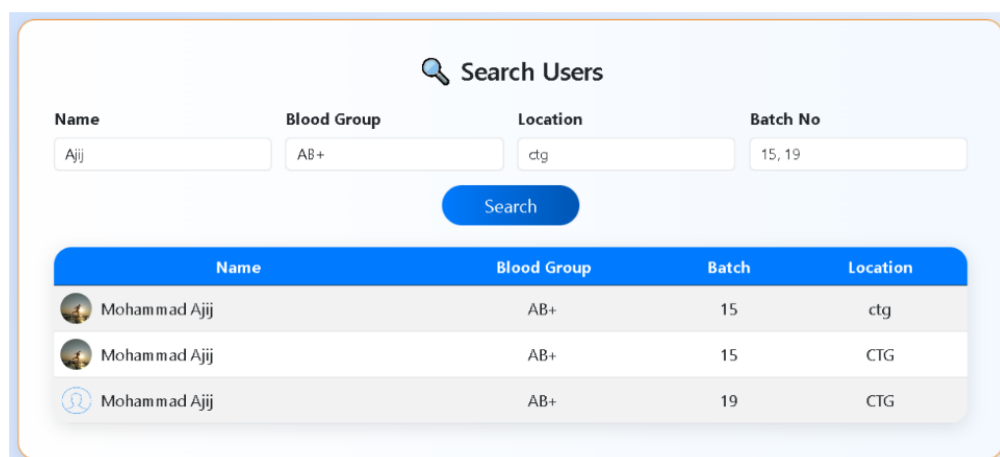


Figure 4.8: Smart User Finder Module

4.1.8 Note Sharing Page

These pages allow users to share their academic notes based on term and subject, upload files, and manage their academic syllabus entries efficiently.

4.1.8.1 Semester Selection Interface

This page allows users to choose their academic term for which they want to share notes. It acts as an entry point to categorize notes by batch and term.

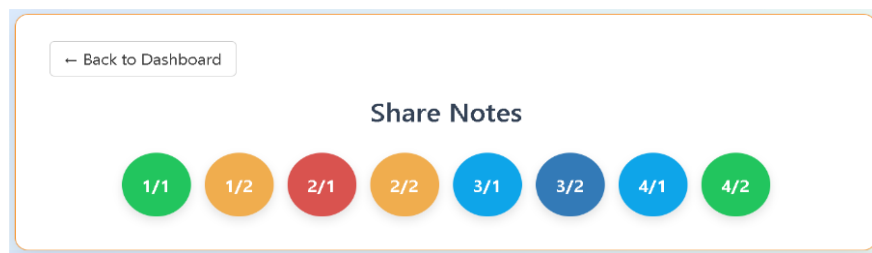


Figure 4.9: Semester Selection Page for Note Share Panel

4.1.8.2 Note Upload Page:

After selecting a term, users can upload notes by choosing a course, entering a title, selecting the note type, and uploading a file. Multiple entries can be added before final submission.

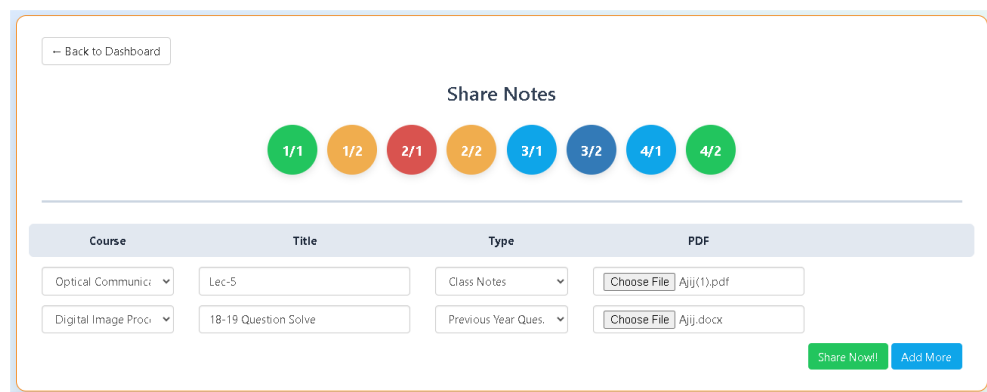
The image shows a web interface for uploading notes. At the top left is a button labeled "← Back to Dashboard". In the center, the text "Share Notes" is displayed above a row of eight colored circular buttons. Below these is a table with columns: Course, Title, Type, and PDF. The table contains two rows of data. The first row shows "Optical Communic" for Course, "Lec-5" for Title, "Class Notes" for Type, and a file "Ajj(1).pdf" for PDF. The second row shows "Digital Image Proc." for Course, "18-19 Question Solve" for Title, "Previous Year Ques." for Type, and a file "Ajj.docx" for PDF. At the bottom right are buttons for "Share Now!!" and "Add More".

Figure 4.10: Note Upload Page

4.1.9 Study Materials Explorer

This section lets users easily browse academic notes by semester, course, and type for efficient study.

4.1.9.1 Semester and Course Display Page

Users first choose a semester, and based on that selection, the system displays the corresponding list of courses available for note browsing.

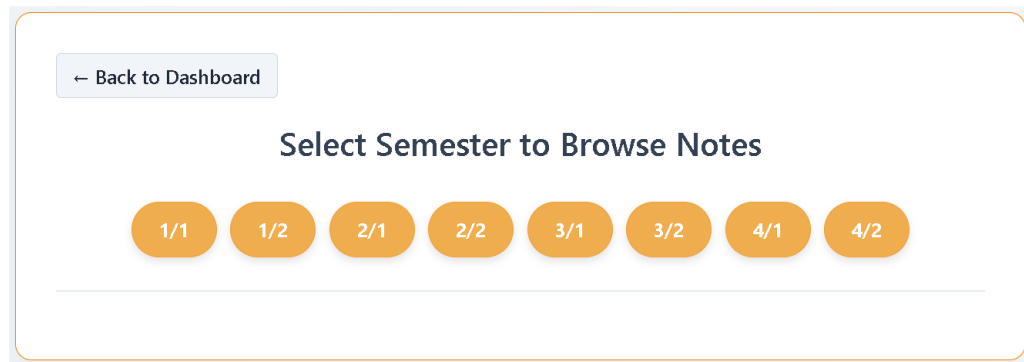


Figure 4.11: Semester and Course Display Page

4.1.9.2 Course & Note Selection

Users can view available courses by semester and select specific note types for each course.

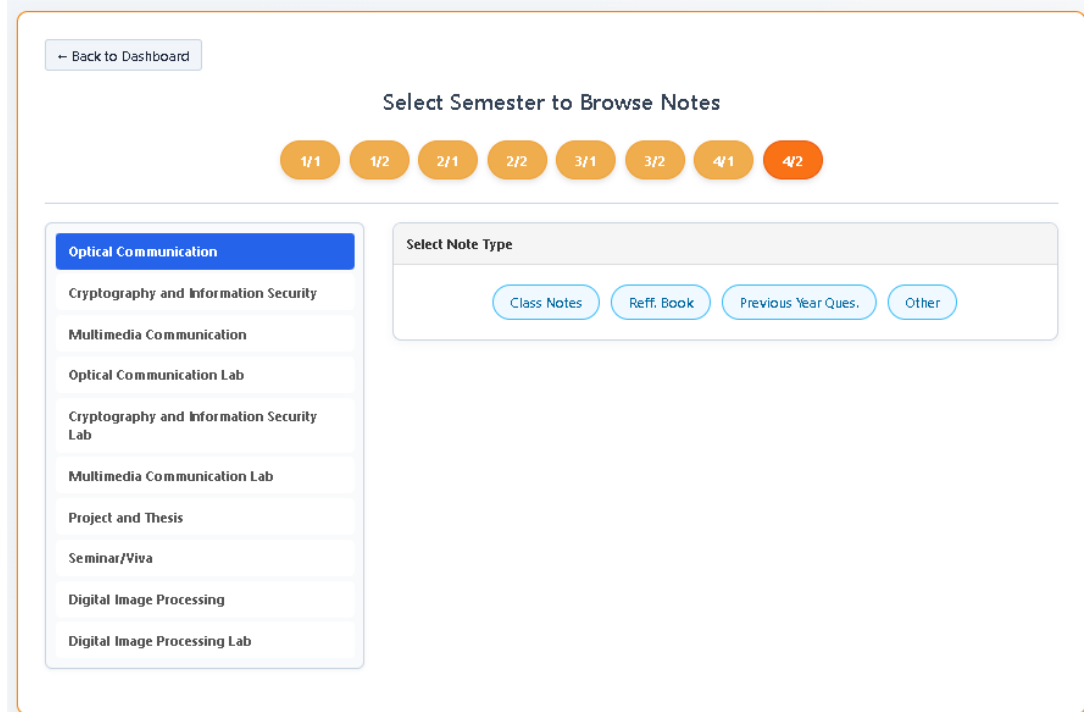


Figure 4.12: Course and Note Type Selection Page

4.1.9.3 Available Notes Viewer

Users view and manage uploaded notes, with options to view, save, or delete notes shared by others.

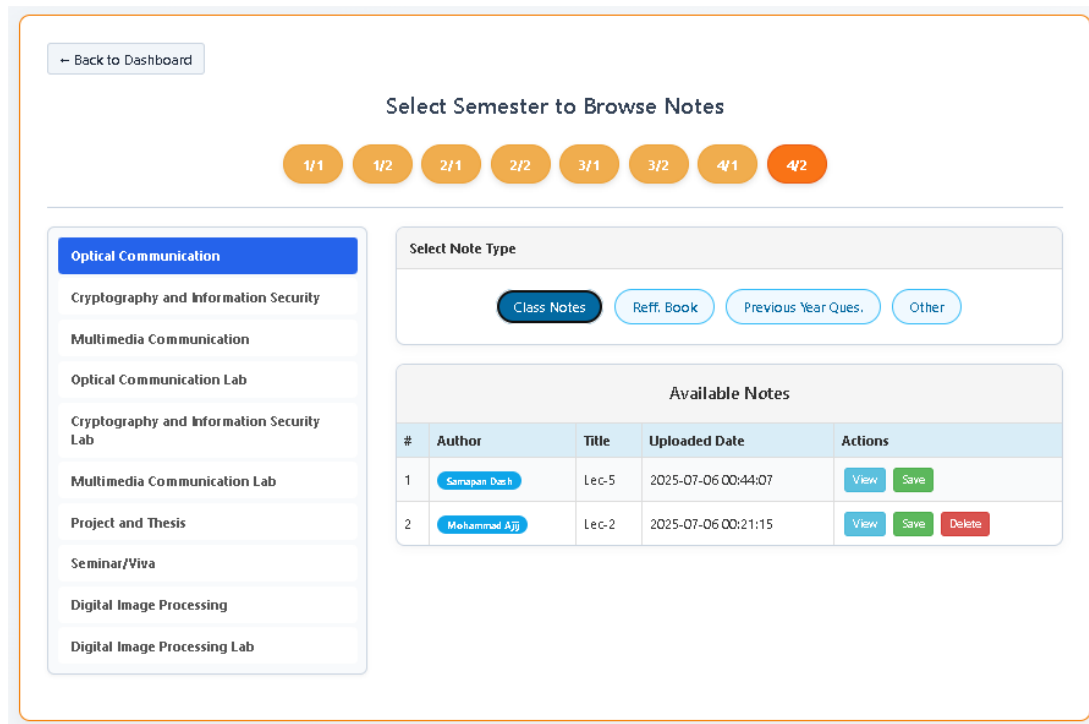


Figure 4.13: Available Notes Viewer Page

4.1.9.4 Latest Uploaded Notes

Displays the most recently shared academic notes for quick access and reference.

📌 Latest Uploaded Notes							
#	Semester	Course	Type	Title	Author	Uploaded	Action
1	4/2	Optical Communication	Class Notes	Lec-6	Mohammad Ajij	2025-07-14 01:49:42	View
2	4/2	Optical Communication	Class Notes	Lec-5	Israt Jahan Tarifa	2025-07-09 22:02:01	View
3	3/2	Digital Signal Processing	Class Notes	Lec-5	Samapan Dash	2025-07-06 00:44:07	View
4	2/1	Digital Logic Design	Other	Lec-2	Mohammad Ajij	2025-07-06 00:21:15	View
5	3/1	Microwave Engineering	Reff. Book	Chapter 2	Mohammad Ajij	2025-06-29 01:03:02	View

Figure 4.14: Latest Uploaded Notes

4.1.10 Group Management Hub

Users can explore, join, create, and manage groups by searching existing ones or defining new groups with selected members and cover images. Admins can handle join requests, assign members, and users can easily access groups they belong to or manage.

4.1.10.1 Group Directory Page

Explore and join available groups on the platform. Users can search for groups, see their membership status, and request to join public groups.

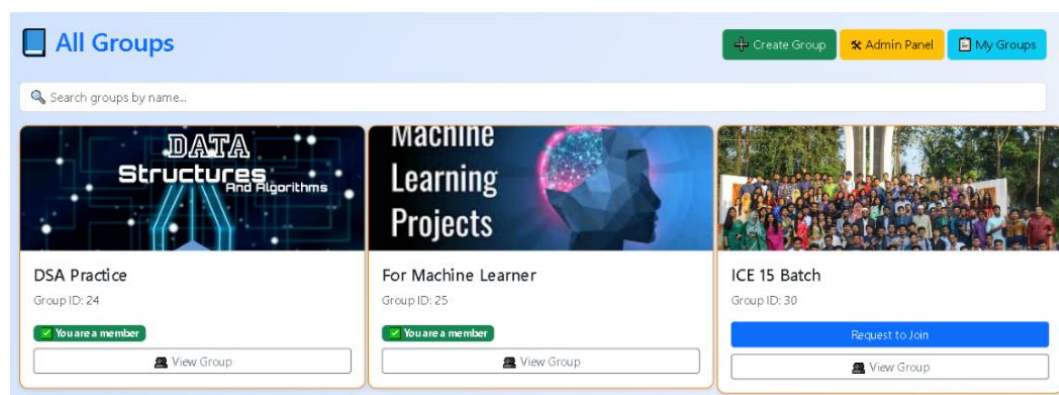


Figure 4.15: Group Directory Page

4.1.10.2 Group Admin Panel

Manage group join requests and manually add users to groups. Admins can accept join requests and assign multiple users to selected groups efficiently. This ensures controlled group membership and supports better collaboration within academic circles.

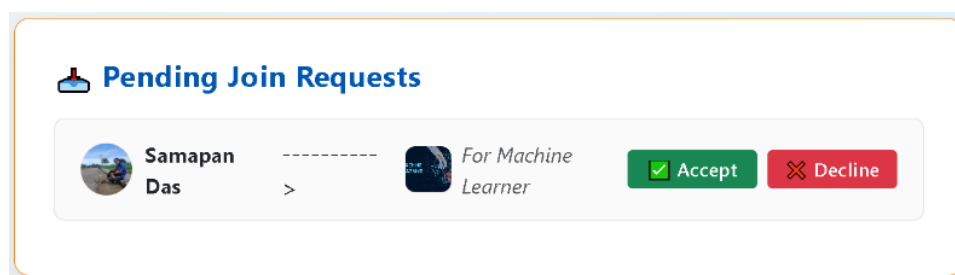
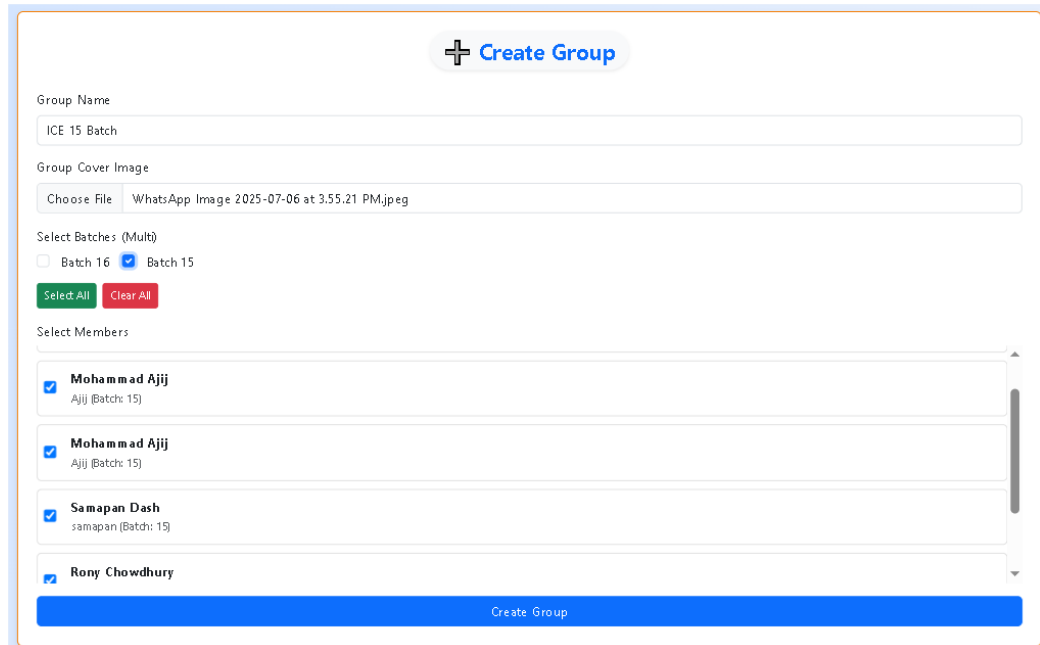


Figure 4.16: Group Admin Panel

4.1.10.3 Group Creation Interface

Create a new group and assign members. Users can define a group name, upload a cover image, select batches, and add members before creating the group.

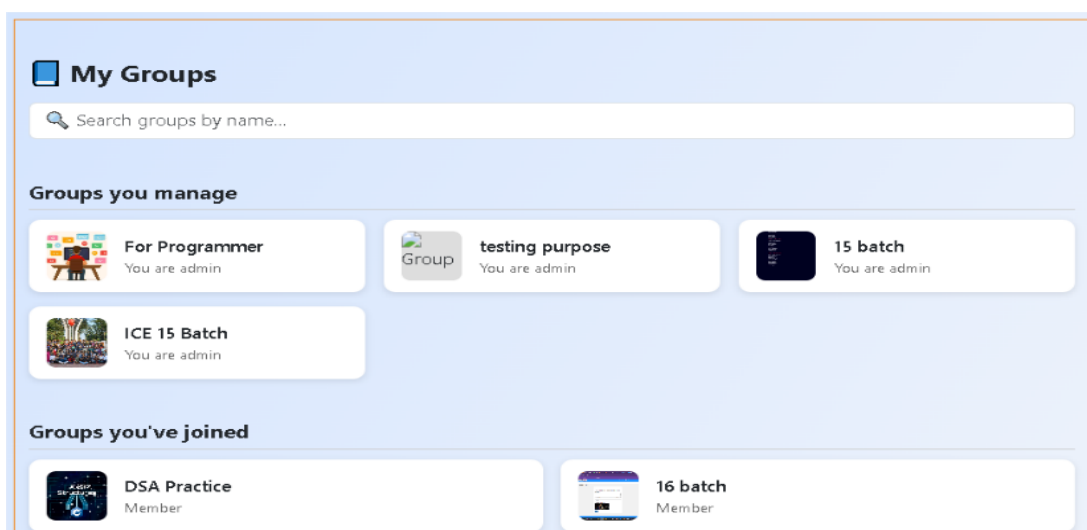


The Group Creation Interface is a web form with a light blue header containing a '+ Create Group' button. Below the header, the form is organized into several sections: 'Group Name' with a text input field containing 'ICE 15 Batch'; 'Group Cover Image' with a 'Choose File' button and a preview of 'WhatsApp Image 2025-07-06 at 3:55:21 PM.jpeg'; 'Select Batches (Multi)' with radio buttons for 'Batch 16' and 'Batch 15' (selected), and 'Select All'/'Clear All' buttons; and 'Select Members' with a scrollable list of members: 'Mohammad Ajij' (Batch: 15), 'Mohammad Ajij' (Batch: 15), 'Samapan Dash' (Batch: 15), and 'Rony Chowdhury'. A blue 'Create Group' button is at the bottom.

Figure 4.17: Group Creation Interface

4.1.10.4 My Groups Page

View and access groups you manage or belong to. This section allows users to quickly browse all their groups, distinguishing between admin roles and membership.



The My Groups Page has a light blue header with a 'My Groups' title and a search bar labeled 'Search groups by name...'. The main content is divided into two sections: 'Groups you manage' and 'Groups you've joined'. The 'Groups you manage' section contains four group cards: 'For Programmer' (You are admin), 'testing purpose' (You are admin), '15 batch' (You are admin), and 'ICE 15 Batch' (You are admin). The 'Groups you've joined' section contains two group cards: 'DSA Practice' (Member) and '16 batch' (Member).

Figure 4.18: My Groups Page

4.1.10.5 Collaborative Group Forum

Groups are used instead of public posts to create a more targeted and structured environment for academic collaboration. They allow users to share content, ask questions, and engage in discussions with a specific set of members, ensuring that information remains relevant and context-specific.

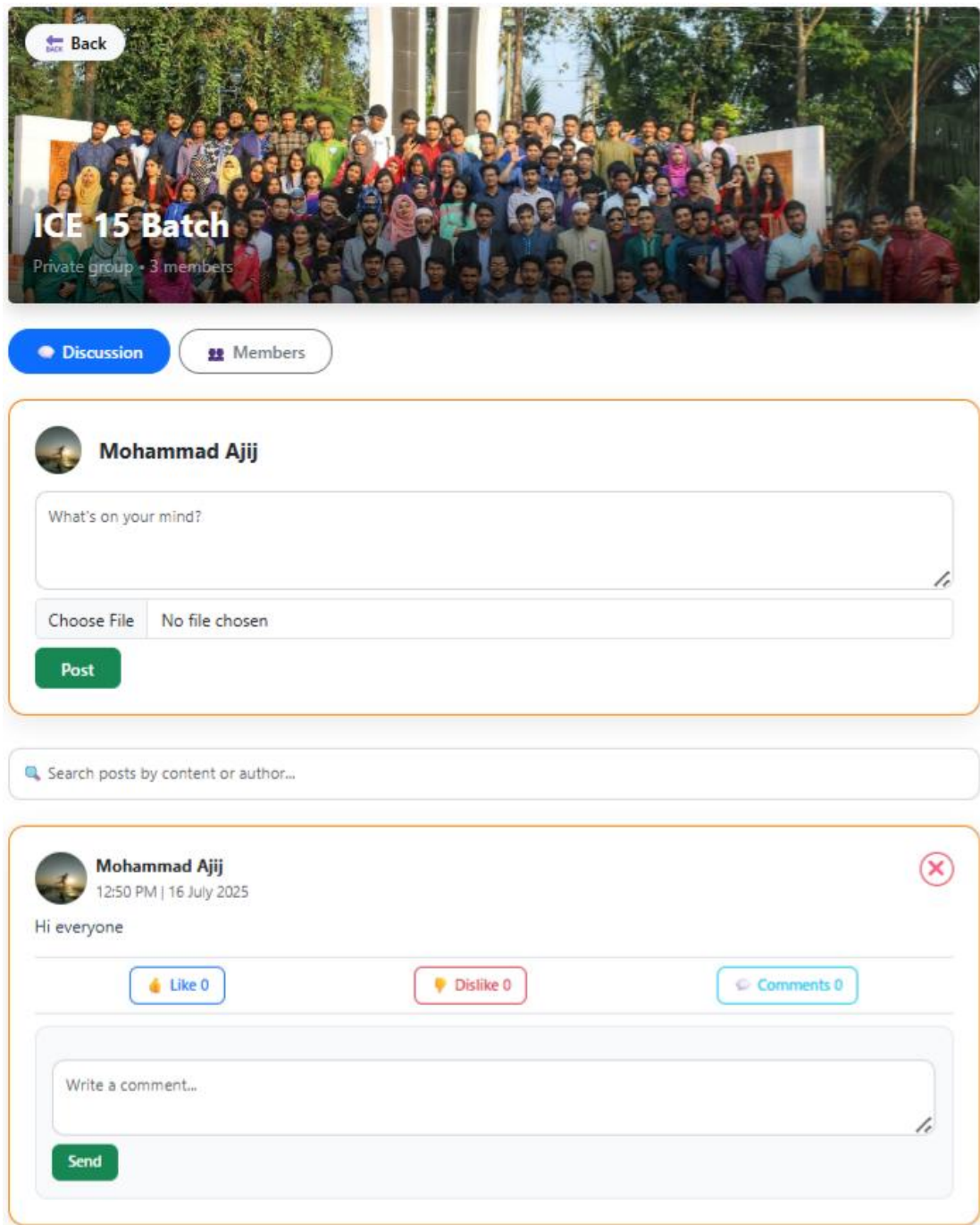
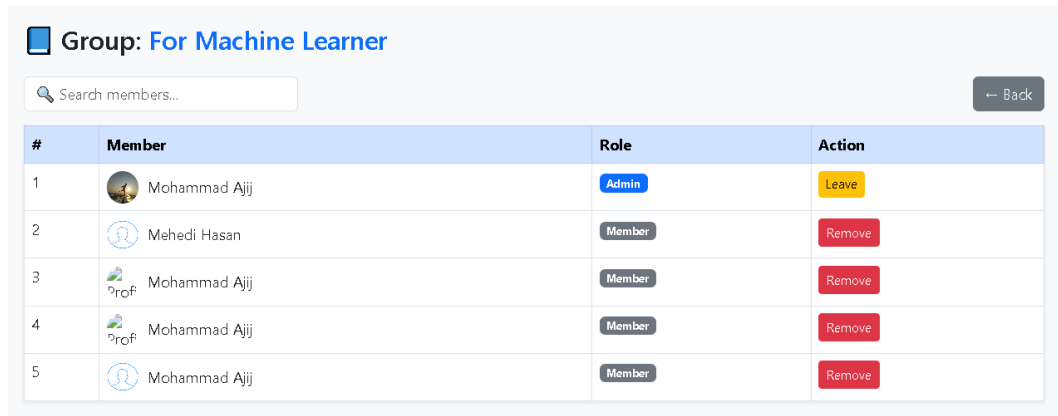


Figure 4.19: Collaborative Group Forum

4.1.10.6 Group Member Management

Displays group members with assigned roles. Admins have the authority to remove members, while users may remove themselves.



#	Member	Role	Action
1	 Mohammad Ajj	Admin	Leave
2	 Mehedi Hasan	Member	Remove
3	 Mohammad Ajj	Member	Remove
4	 Mohammad Ajj	Member	Remove
5	 Mohammad Ajj	Member	Remove

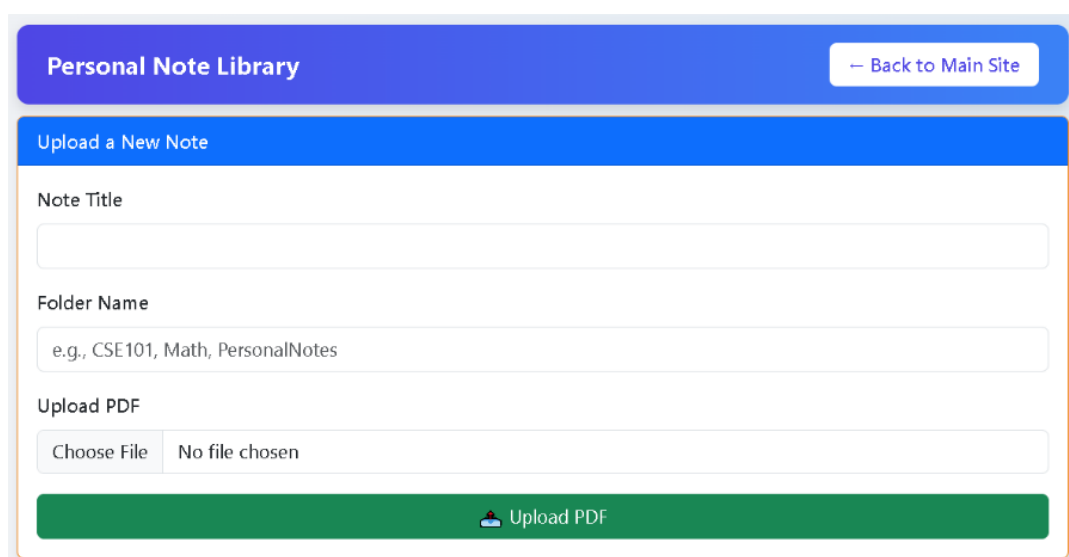
Figure 4.20: Group Member Management

4.1.11 Private Notes Archive

A secure space where users can upload, organize, and manage their personal study materials for easy access and future reference.

4.1.11.1 Note Upload Interface

Enables users to upload and organize notes by title, folder, and file.



Personal Note Library [← Back to Main Site](#)

Upload a New Note

Note Title

Folder Name

Upload PDF
 No file chosen

Figure 4.22: Note Upload Interface

4.1.11.2 Organized Note Management

Allows users to view, edit, or delete uploaded notes sorted by folders for easy access.

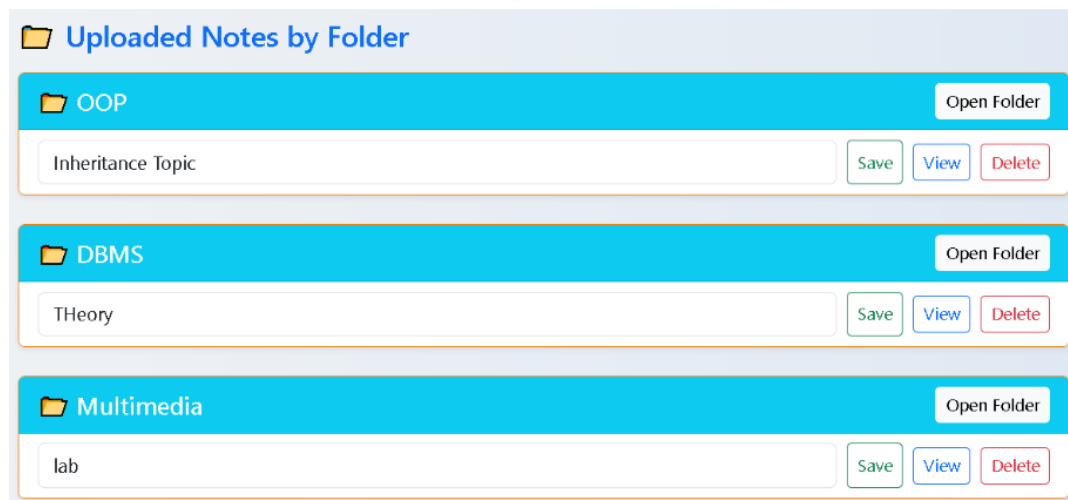


Figure 4.23: Organized Note Management Panel

4.1.12 Smart Task Tracker

A simple module to add, manage, and track daily academic or personal tasks with completion status.

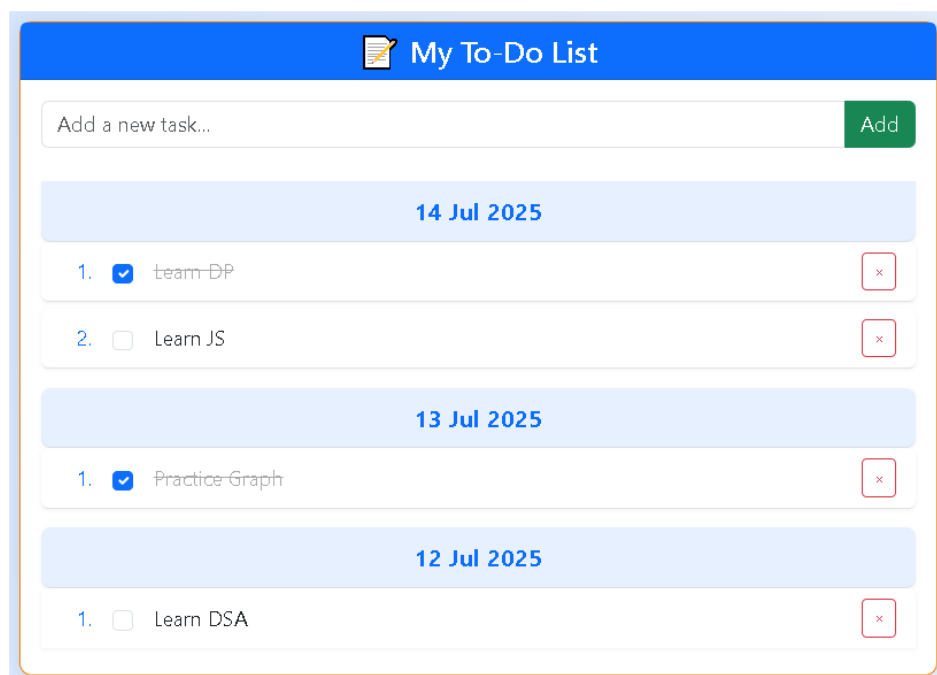


Figure 4.24: Smart Task Tracker Panel

4.1.13 Notification Panel

Displays real-time updates for new class schedules, shared posts, and uploaded notes from users.

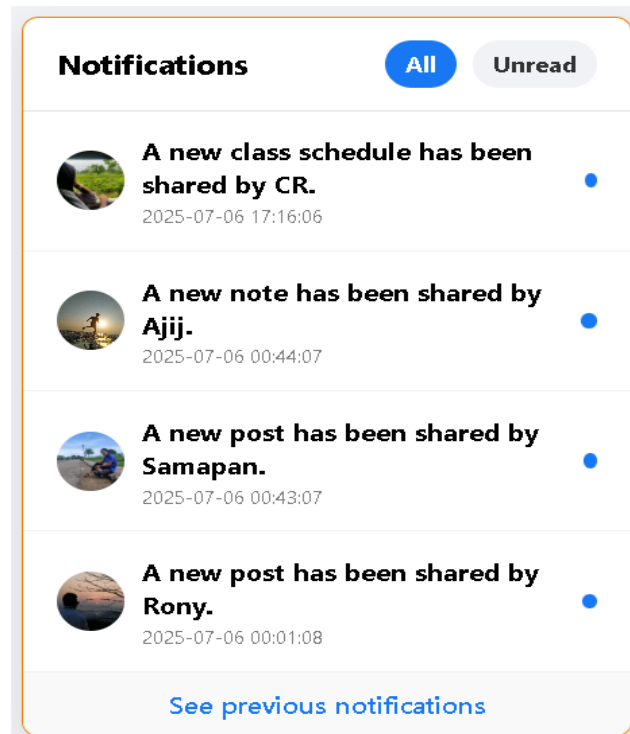


Figure 4.25: Notification Panel

4.1.14 Unified Profile Hub

A centralized interface to view, update, and manage your personal profile, notes, and task activities efficiently.

4.1.14.1 User Dashboard

This dashboard provides a user-friendly overview of the student's profile, displaying essential details such as name, registration number, contact information, blood group, and batch. It offers easy navigation to key features, including account editing. Additionally, the "My To-Do List" section helps in tracking and managing daily tasks, promoting better time management and productivity within the learning environment.

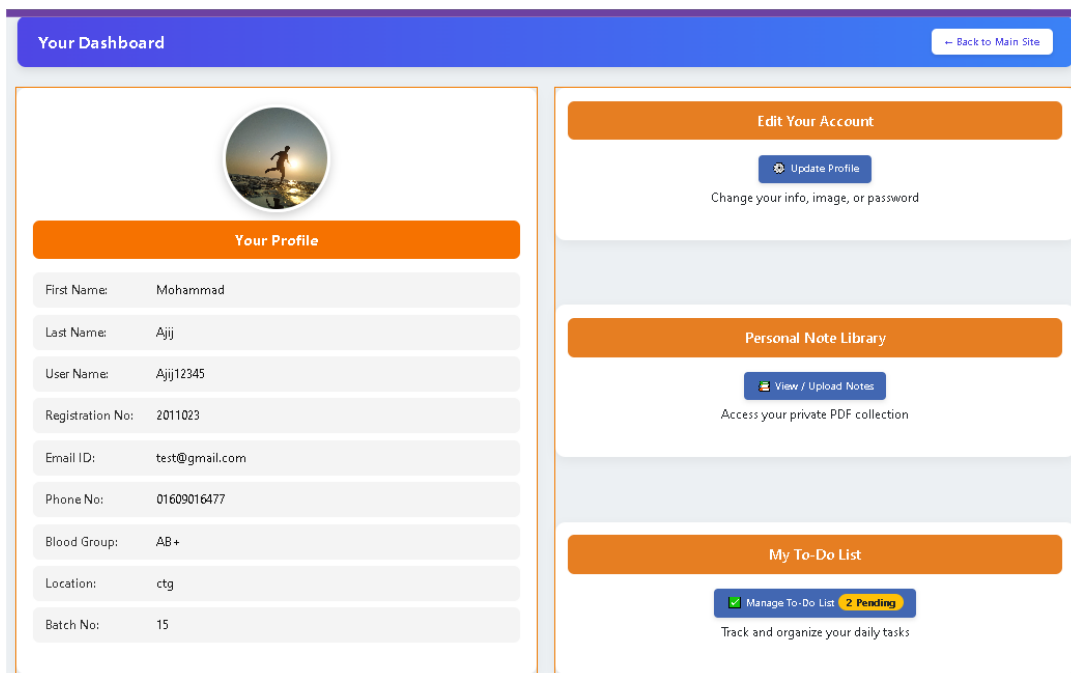


Figure 4.26: User Dashboard Page

4.1.14.2 Profile Edit Interface

Allows users to update their personal information.

Edit Your Account

First Name

Mohammad

Last Name

Ajij

User Name

Ajij12345

Email

test@gmail.com

Phone

01609016477

Registration No

2011023

Blood Group

AB+

Location

ctg

Batch No

15

Update Account

Figure 4.27: Profile Edit Interface

4.1.15 View User Profile

Displays detailed information of a selected user, including name, contact, academic details, and location for administrative or personal reference. The clean and simple layout ensures quick access to essential information such as registration number, batch, blood group, and email. This profile view helps teachers, administrators, or fellow students verify user identity and communicate as needed. The “Go Back” button allows smooth navigation, enhancing the overall user experience within the platform.

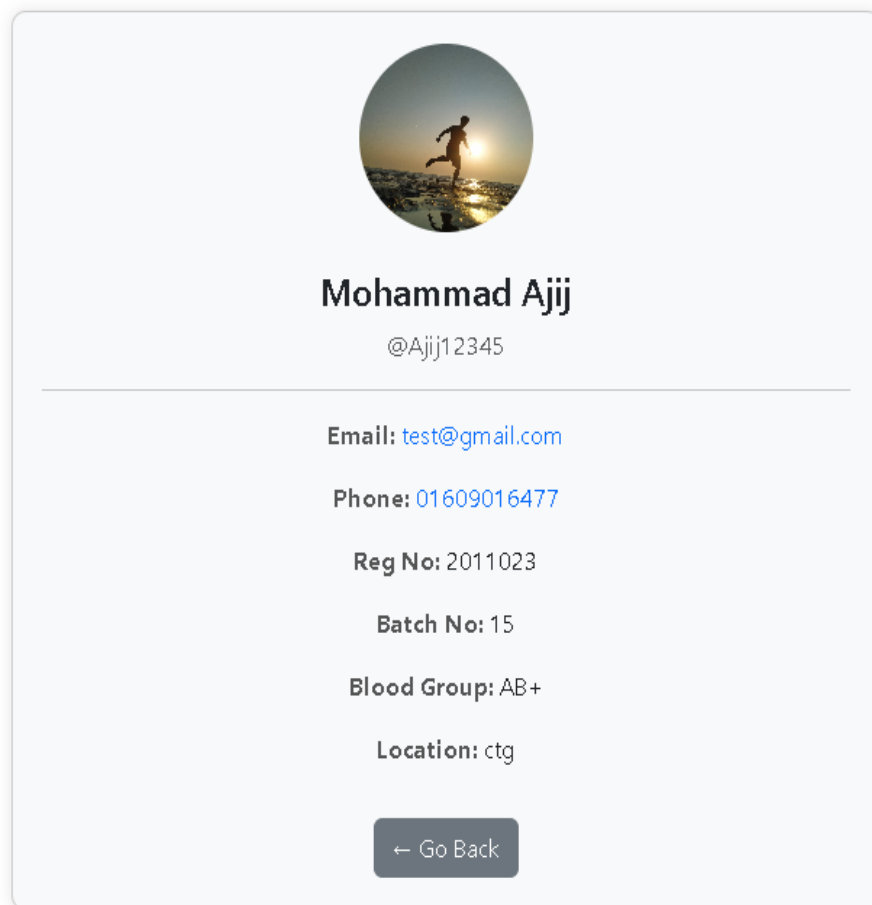


Figure 4.28: View User Profile

4.2 Admin Module

The Admin Module streamlines academic administration by allowing admins to approve users, manage schedules, publish notices, and maintain syllabus records—all from a unified interface.

4.2.1 Admin Dashboard

Allows verified admins to manage users, update syllabus, publish notices, and maintain student records from a central dashboard.

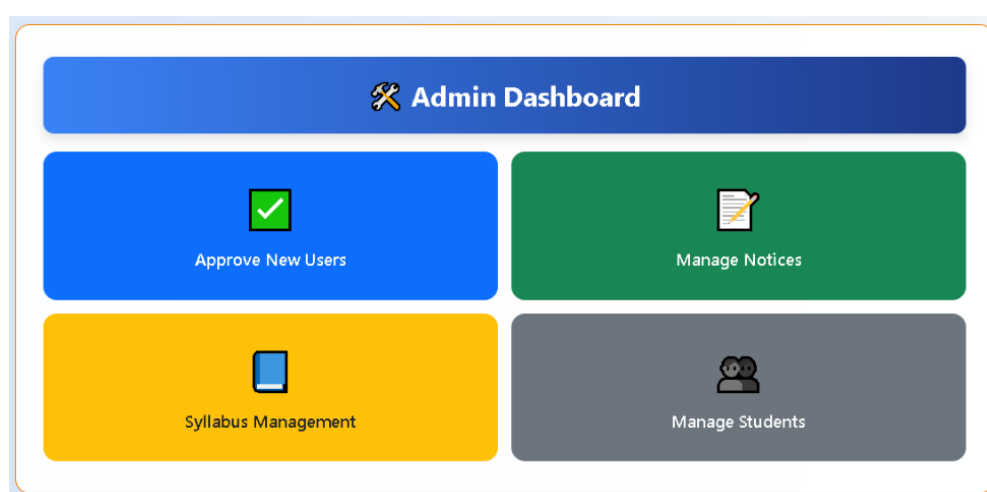


Figure 4.29: Admin Dashboard Page

4.2.2 Class Schedule Management

Admins can add or edit class schedules by selecting semester, batch, and date. It ensures smooth planning and quick updates for any changes in timing, venue, or faculty.

4.2.2.1 Update Class Schedule

Allows admins to create and assign class schedules by selecting semester, batch, date, time, and venue—ensuring clear, conflict-free planning and organized batch-wise routines. The feature also simplifies updates to existing schedules, improving overall timetable management.

[Back](#)

Update Class Schedule

Select Semester:

2/1

Select Batch:

18

Select Date:

07/14/2025

Course	From	To	Venue	Select
Digital Logic Design	10 : 0	11 : 0	802	☒
Digital Logic Design Lab	14 : 0	15 : 0	805	☒

Submit

Figure 4.30: Update Class Schedule Page

4.2.2.2 Edit Class Schedule Panel

Admins can view and update schedules by batch and date to handle changes and avoid conflicts.

[Edit Class Schedule](#)

Select Batch:

15

Select Date:

07/16/2025

Course	From	To	Venue	Action
Optical Communication	2 : 0	2 : 10	802	Delete
Multimedia Communication	10 : 0	10 : 30	802	Delete

Update Schedule

Figure 4.31: Class Schedule Editing Panel

4.2.3 Notice Management Panel

Enables admins to upload, view, and delete official notices. The system supports PDF uploads and displays all notice records in a manageable table format. Each notice includes details such as type, title, and upload date, helping ensure timely and organized communication with students and staff. With a clear layout, this feature enhances administrative efficiency and keeps important information accessible.

 **Upload New Notice**

[← Back to Dashboard](#)

Type

Title

Upload PDF

Choose FileNo file chosen

Upload Notice

 **All Uploaded Notices**

#	Type	Title	Uploaded Date	Actions
1	Exam form fillup	Exam	2025-07-06	ViewSaveDelete
2	Re Admission 2023-24	Re Admission	2025-06-19	ViewSaveDelete
3	Re Admission 2023-24	Re Admission	2025-06-19	ViewSaveDelete

Figure 4.32: Notice Management Panel

4.2.4 Exam Schedule Management

The Exam Schedule Management feature provides admins with control over scheduling, editing, and deleting exam entries based on date, exam name, and batch. This functionality supports efficient academic planning and timely communication with students. Through the Edit Exam Schedule Panel, admins can make quick modifications or deletions, ensuring that any changes or rescheduling are accurately reflected across the system, enhancing overall academic preparedness.

4.2.4.1 Update Exam Schedule

Admins can schedule exams by selecting date, entering exam name, and assigning a batch—supporting organized planning and timely communication. This panel streamlines the process of managing multiple upcoming exams, allowing easy additions and updates for efficient academic coordination.

Date	Exam Name	Batch No
07/24/2025	DSA Lab Exam	16
07/31/2025	Defense	15

Figure 4.33: Update Exam Schedule Page

4.2.4.2 Edit Exam Schedule Panel

Admins can quickly view, update, or delete scheduled exams for a batch, ensuring accurate and up-to-date exam planning. It provides a user-friendly interface to ensure timely scheduling and organization of exams for effective academic planning.

Select Batch: 15

Date	Exam Name	ID	Action
07/31/2025	Defense	38	Delete

Figure 4.34: Exam Schedule Editing Panel

4.2.5 Academic Syllabus Management

A centralized interface for managing course syllabus entries—enabling admins to add new courses, filter by year-term, and update or delete existing course records efficiently.

Academic Syllabus Management [← Back to Dashboard](#)

SI No. Course Code Course Title Credit Course Type

Year-Term

[Add Syllabus](#)

Filter by Year-Term

CSV Print PDF

SI No.	Course Code	Course Title	Credit	Type	Year-Term	Actions
1	ICE 4101	Artificial Intelligence and Neural Networks	3	Compulsory	Year-4 Term-I	Edit Delete
1	ICE 4201	Optical Communication	3	Compulsory	Year-4 Term-II	Edit Delete
2	ICE 2103	Object Oriented Programming (C++)	3	Compulsory	Year-2 Term-I	Edit Delete

Showing 1 to 3 of 22 entries

Previous [1](#) [2](#) [3](#) [4](#) [5](#) ... [8](#) Next

Figure 4.35: Syllabus Management Panel

4.2.6 Approve New Users

Allows the admin to review newly registered users and either approve or cancel their access, ensuring controlled user management. This helps maintain platform security and prevents unauthorized access.

Approve New Users [← Back to Dashboard](#)

ID	Username	Full Name	Email	Actions
19	samapan	Samapan Das	samapan@gmail.com	Approve Cancel
20	test99	Mohammad Ajij	mohammadajij120386@gmail.com	Approve Cancel

Figure 4.36: User Management Panel

4.3 Teacher Module

Enables teachers to manage class schedules, exams, and academic resources. It includes registration, login, a dashboard with panels for updates, resource sharing, and easy schedule tracking streamlining teaching and communication.

4.3.1 Home Page

The dashboard appears when a teacher visits the platform's main URL.

4.3.1.1 Main Dashboard Sections

The Main Dashboard Consists of Five Key Sections for Streamlined Academic Management:

i. Class Schedule Panel

Helps teachers manage time and stay prepared with a clear view of daily tasks. Also assists in avoiding schedule conflicts and improving class preparedness.

ii. Newsfeed Panel

Enables teachers to share updates and resources for academic engagement. Facilitates interaction between teachers and students through posts and comments.

iii. Exam Panel

Centralizes exam schedules for effective planning and student coordination. Keeps both teachers and students aware of upcoming assessments at a glance.

iv. Resource Sharing Panel

Encourages resource sharing and academic support by enabling teachers to distribute lecture materials and assignments directly to students.

v. Campus View Panel

Displays a featured image of the campus to provide a visual connection with the institution. Adds a welcoming and familiar touch to the user interface.

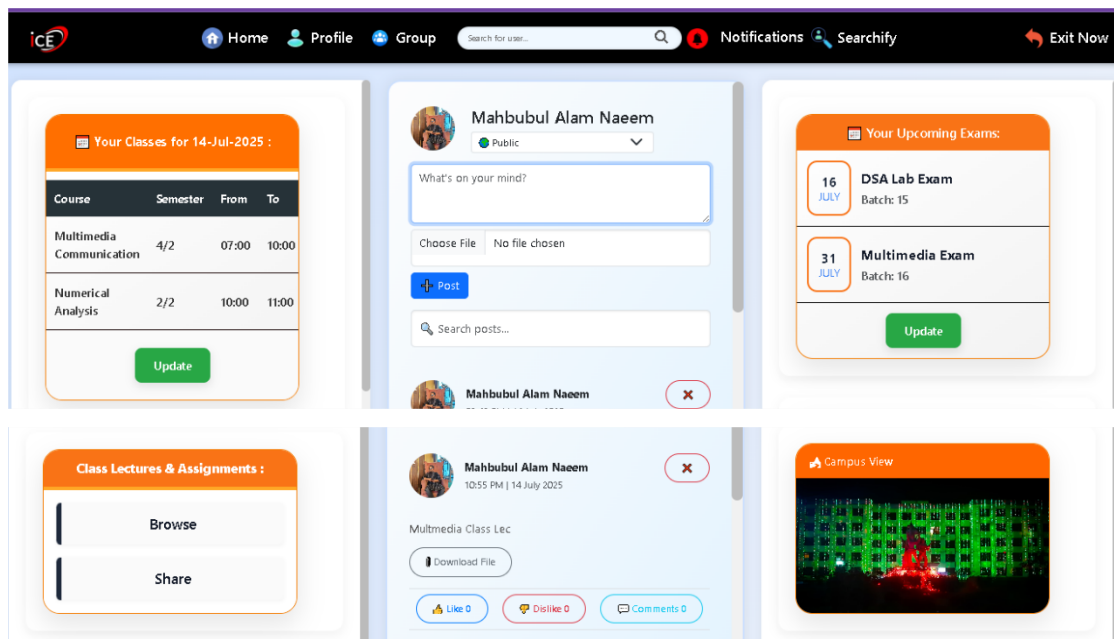


Figure 4.37: Home Page

4.3.2 Login Page

After registration, Teacher can login their account by inserting valid email address and password. The system verifies the credentials to grant secure access to the platform.

Figure 4.38: Login Page

4.3.3 Registration Page

Teachers can register by submitting their details. After admin approval, they can access schedules, share notes, and collaborate academically.

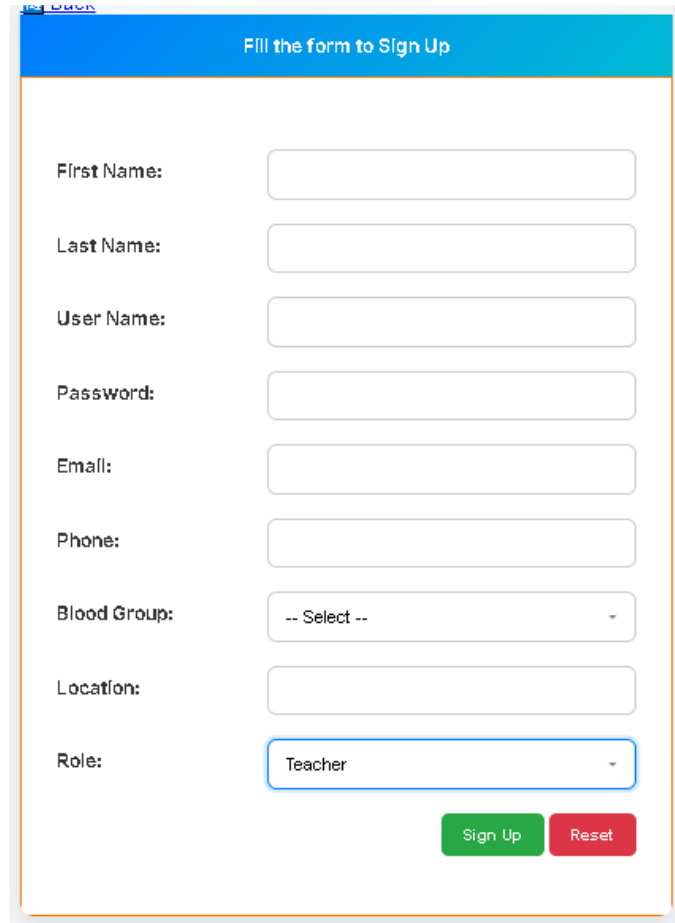
The image shows a web-based registration form. At the top, there is a blue header bar with the text "Fill the form to Sign Up" in white. Below the header, the form is enclosed in a light gray border. It contains several input fields: "First Name:", "Last Name:", "User Name:", "Password:", "Email:", "Phone:", "Blood Group:" (with a dropdown menu showing "-- Select --"), "Location:", and "Role:" (with a dropdown menu showing "Teacher"). At the bottom right of the form, there are two buttons: a green "Sign Up" button and a red "Reset" button.

Figure 4.39: Registration Page

4.3.4 Class Schedule Management

Allows teachers to set, update, and view their daily class routines—ensuring accurate scheduling, better time management, and timely communication with students.

4.3.4.1 Update Class Schedule

Enables teachers to set or update their class times, dates, and batches. This feature is crucial for maintaining accurate routines and informing students of any changes.

Update Class Schedule

Select Semester: 2/1

Select Batch: 18

Select Date: 07/14/2025

Course	From	To	Venue	Select
Digital Logic Design	10:00	11:00	802	☒
Digital Logic Design Lab	14:00	15:00	805	☒

Submit

Figure 4.40: Update Class Schedule Page

4.3.4.2 View Daily Teaching Schedule

Displays the teacher's daily class schedule to help manage time and avoid overlaps. It ensures teachers are always prepared for their sessions and maintain punctuality.

Your Classes for 14-Jul-2025 :

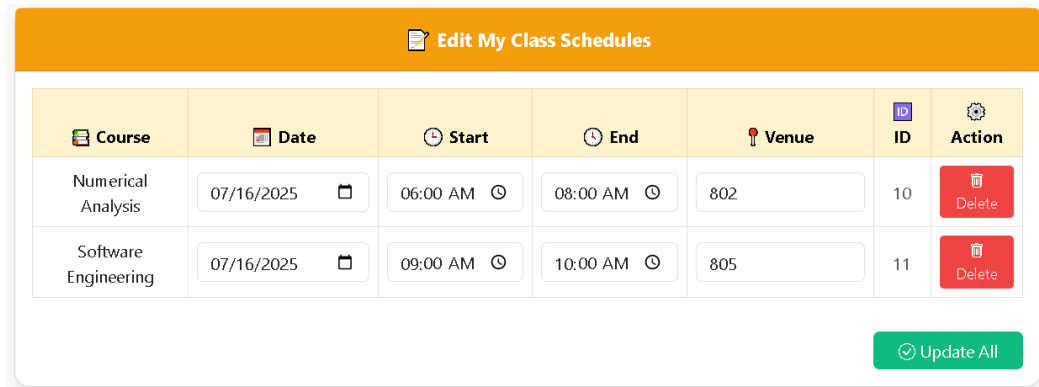
Course	Semester	From	To
Software Engineering	4/1	09:10	09:50
Optical Communication	4/2	11:40	12:00
Digital Logic Design	2/1	14:00	14:00

Update

Figure 4.41: View Daily Teaching Schedule

4.3.4.3 Class Schedule Editing Panel

Allows teachers to update or delete class schedules, ensuring accurate and timely coordination with students.



The 'Edit My Class Schedules' panel features a table with columns for Course, Date, Start, End, Venue, ID, and Action. It lists two classes: Numerical Analysis and Software Engineering, both on 07/16/2025. Each class has a 'Delete' button. An 'Update All' button is at the bottom right.

Course	Date	Start	End	Venue	ID	Action
Numerical Analysis	07/16/2025	06:00 AM	08:00 AM	802	10	Delete
Software Engineering	07/16/2025	09:00 AM	10:00 AM	805	11	Delete

Update All

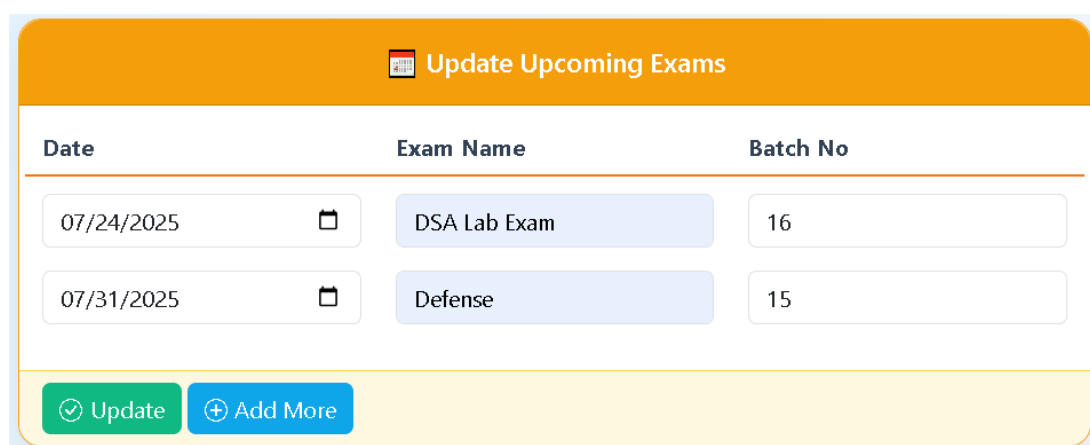
Figure 4.42: Class Schedule Editing Panel

4.3.5 Exam Schedule Management

Helps teachers view, add, and update exam schedules for different batches—ensuring smooth coordination, avoiding conflicts, and supporting effective academic planning.

4.3.5.1 Update Exam Schedule

Allows teachers to add or change exam schedules easily. It helps in coordinating assessments, communicating with students, and maintaining a structured academic calendar.



The 'Update Upcoming Exams' panel displays a table with columns for Date, Exam Name, and Batch No. It lists two exams: DSA Lab Exam and Defense. Below the table are 'Update' and 'Add More' buttons.

Date	Exam Name	Batch No
07/24/2025	DSA Lab Exam	16
07/31/2025	Defense	15

Update Add More

Figure 4.43: Exam Management Panel

4.3.5.2 Exam Schedule Editing Panel

This panel enables teachers to edit and manage exam schedules for different batches efficiently. It ensures accurate exam planning, avoiding overlaps and keeping students and faculty informed about assessment dates.

Date	Exam Name	Batch	Exam ID	Action
07/16/2025	DSA Lab Exam	15	3	Delete
07/31/2025	Multimedia Exam	16	5	Delete

Update All

Figure 4.44: Exam Schedule Editing Panel

4.3.5.3 Track Exam Schedules

This section helps teachers stay updated with upcoming exams for different batches. It ensures proper planning, avoids scheduling conflicts, and keeps teachers aware of their assessment responsibilities.

Your Upcoming Exams:

- 24 JULY** DSA Lab Exam
Batch: 16
- 31 JULY** Defense
Batch: 15

Figure 4.45: Track Exam Schedules

4.4 Visitor Module

The Visitor Module lets users access notices, syllabus, results, and routines without login. It also offers AI academic help and a contact form for support.

4.4.1 Student Dashboard

Allows visitors to preview available academic features such as notices, syllabus, results, class routines, exam routines, and seat plans without requiring login.

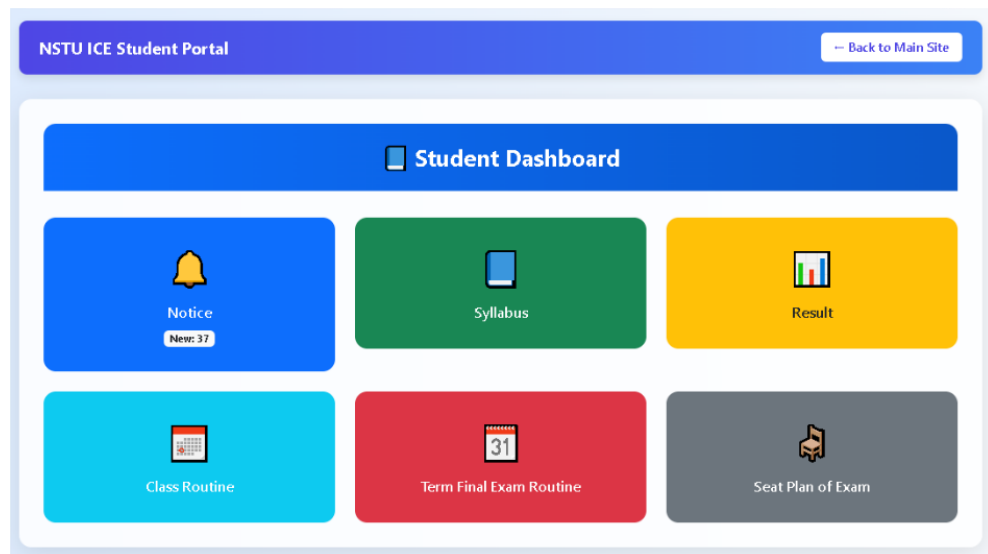


Figure 4.46: Student Dashboard

4.4.2 Notice Board Overview

Displays a list of all uploaded notices with options to view or save each notice for future reference.

#	Type	Title	Uploaded	Actions
1	Exam form fillup	Exam	06 Jul 2025	View Save
2	Re Admission 2023-24	Re Admission	19 Jun 2025	View Save
3	Re Admission 2023-24	Re Admission	19 Jun 2025	View Save

Figure 4.47: Notice Board Overview

4.4.3 Academic Syllabus Overview

View and filter course details by year and term for academic reference.

4.4.3.1 View Academic Syllabus

Displays a complete list of courses with code, credit, and type for a selected academic year and term. Helps users understand the course structure and requirements. It ensures easy access to official curriculum details for better academic planning.

Academic Syllabus					
Filter by Year-Term					
CSV Print PDF					
SI No.	Course Code	Course Title	Credit	Type	Year-Term
1	ICE 4101	Artificial Intelligence and Neural Networks	3	Compulsory	Year-4 Term-I
1	ICE 4201	Optical Communication	3	Compulsory	Year-4 Term-II
2	ICE 4103	Software Engineering	3	Compulsory	Year-4 Term-I
2	ICE 4203	Cryptography and Information Security	3	Compulsory	Year-4 Term-II
3	ICE 4105	Antenna and Satellite Communication	3	Compulsory	Year-4 Term-I
3	ICE 4205	Multimedia Communication	3	Compulsory	Year-4 Term-II
4	ICE 4107	Wireless and Mobile communication	3	Compulsory	Year-4 Term-I
4	ICE 4202	Optical Communication Lab	1	Compulsory	Year-4 Term-II
5	ICE 4109	Simulation and Modeling	3	Compulsory	Year-4 Term-I
5	ICE 4204	Cryptography and Information Security Lab	1	Compulsory	Year-4 Term-II

Showing 1 to 10 of 21 entries

[Previous](#) [1](#) [2](#) [3](#) [Next](#)

Figure 4.48: View Academic Syllabus

4.4.3.2 Filter Syllabus by Term

Allows users to filter and view the syllabus based on selected academic terms, showing detailed course information for efficient academic planning. This feature helps students and teachers quickly access relevant course materials for a specific term. It enhances usability by minimizing clutter and focusing only on the selected academic timeframe. It also supports better preparation by aligning course content with the academic timeline.

SI No.	Course Code	Course Title	Credit	Type	Year-Term
1	ICE 4201	Optical Communication	3	Compulsory	Year-4 Term-II
2	ICE 4203	Cryptography and Information Security	3	Compulsory	Year-4 Term-II
3	ICE 4205	Multimedia Communication	3	Compulsory	Year-4 Term-II
4	ICE 4202	Optical Communication Lab	1	Compulsory	Year-4 Term-II
5	ICE 4204	Cryptography and Information Security Lab	1	Compulsory	Year-4 Term-II
6	ICE 4206	Multimedia Communication Lab	1	Compulsory	Year-4 Term-II
7	ICE 4218	Project and Thesis	4	Compulsory	Year-4 Term-II
8	ICE 4200	Seminar/Viva	1	Compulsory	Year-4 Term-II
9	ICE 4207	Digital Image Processing	3	Optional	Year-4 Term-II
10	ICE 4208	Digital Image Processing Lab	1	Optional	Year-4 Term-II

Showing 1 to 10 of 11 entries (filtered from 21 total entries)

Previous 1 2 Next

Figure 4.49: Filter Syllabus by Term

4.4.4 Academic AI Assistant Chat Module

Offers an interactive chat interface where users can ask academic questions and receive detailed AI-generated responses, such as code explanations and subject-specific help. Supports a wide range of topics including programming, mathematics, and theoretical concepts. Enhances self-learning by providing instant guidance and reducing dependency on external sources.

4.4.4.1 AI Query Panel

Allows users to input academic questions or code-related prompts. Enables students to engage in personalized learning through a conversational interface.

Academic AI Assistant

Send
Reset Chat

Figure 4.50: AI Query Panel

4.4.4.2 AI Response Viewer

Displays the AI-generated answers, including code and explanation

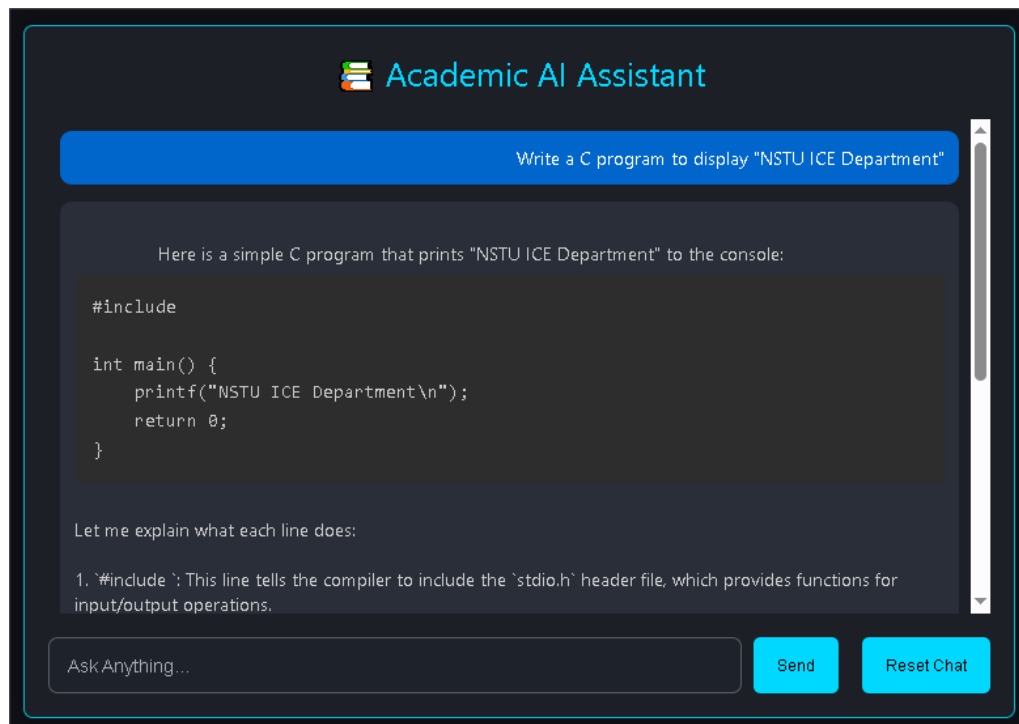


Figure 4.51: AI Response Viewer

4.4.5 Contact & Support Form

Allows visitors to send queries via form or contact through email, phone, or social links for quick support.

The screenshot shows a 'GET IN TOUCH' contact form on a dark background. On the left, under the heading 'For Any Queries and Support', there are three contact options: 'Location' (Kutubdia, Cox's Bazar), 'Write to us at' (mohammadajij120386@gmail.com), and 'Call us on' (+880 1609016477). Below these are icons for LinkedIn, Facebook, Instagram, and Twitter. On the right, there are four input fields: 'Your Name', 'Your Email', 'Subject', and a larger 'Type Your Message Here...' field. A 'Send Message' button is at the bottom right.

Figure 4.52: Contact & Support Form

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

The Web-Based Platform for Academic Discussions and Resource Sharing aims to revolutionize the way students interact and collaborate academically. By offering a range of features that blend social networking functionalities with educational tools, this system promotes efficient knowledge sharing, time management, and academic organization. It enables students to engage in focused discussions, share resources, manage schedules, and receive personalized study tips, all within a distraction-free environment. Ultimately, the platform will replace many manual academic processes with a more efficient, digitalized approach, benefiting both students and faculty by saving time and enhancing communication.

5.2 Future Work

For the future of this project more aspects will be included to expand on all the already present features that help the academic setting. There is still room for improvements though, such as integrating a mobile application, to give the students the ability to use the platform even if not on the computer. Furthermore, there is the ability to offer even more specific using higher levels of Artificial Intelligence incorporated to the system. The system can also be extended to the other departments or institutions that would enhance its coverage area. Future work will also consist in guaranteeing that the platform advances in terms of scalability and security while answering the novel demands of academics and including new technologies.

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